

Rubinstein–Sarnak densities for weighted prime number races

Raghav Sharma*

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Abstract

For $m \in \mathbb{R}$ consider the weighted prime race $D_m(x) = \sum_{p \leq x, p \equiv 3 \pmod{4}} p^m - \sum_{p \leq x, p \equiv 1 \pmod{4}} p^m$ and its analogues for other moduli. The regime $m \leq 0$ has a developed literature: the critical $\frac{1}{2} \log \log x$ deflection at $m = -\frac{1}{2}$ (Aoki–Koyama; Sheth), the sign-change threshold (Shimada–Koyama), and, for $m < -\frac{1}{2}$, limits expressible through Meissel–Mertens constants for progressions (Languasco–Zaccagnini). We contribute the pieces this literature does not contain: (i) the first numerical Rubinstein–Sarnak logarithmic densities for a weighted race between primes in arithmetic progressions (for weighted prime-sum races against main terms, cf. Lamzouri’s Mertens-race density [13]), computed across the delicate range under the standard GRH + LI hypotheses by deterministic Fourier inversion — e.g. for modulus 4, $\delta(1) = 0.79763$, $\delta(2) = 0.69452$, $\delta(3) = 0.64585$, with the calibration $\delta(0) = 0.995928$ recovering the classical Rubinstein–Sarnak value to six decimals; (ii) the growing-weight regime $m > 0$ (we are not aware of prior Rubinstein–Sarnak density computations or weight-aspect density asymptotics for growing weights p^m , $m > 0$), with exact sieve computations to 10^{10} agreeing with the predicted densities to $\lesssim 4 \times 10^{-3}$ for modulus 4; (iii) an asymptotic dissolution law in the weight aspect, $\delta_q(m) - \frac{1}{2} \sim (2\pi \cdot 2(m + \frac{1}{2}) \log \frac{q(m+1/2)}{2\pi})^{-1/2}$, an analogue of the Fiorilli–Martin density asymptotics, proved here with an error term for the density of the limiting distribution of any real primitive character under GRH and $L(\frac{1}{2}, \chi) \neq 0$ — under LI it applies to the logarithmic density of the mod-4 race itself — and verified numerically in moduli 3 and 4. We complement the computations with an interpolation theorem, proved by the framework of Akbary–Ng–Shahabi and the method of Humphries: under GRH and LI for $L(s, \chi_4)$, for every $m \in (-\frac{1}{2}, \infty)$ the logarithmic density $\delta(m)$ of the set of x where the class 3 (mod 4) leads exists and lies in $(\frac{1}{2}, 1)$, the map $m \mapsto \delta(m)$ is continuous, and $1 - \delta(m) \leq \exp(-1/(2C(2m+1)^2))$ with $C = \sum_{\gamma > 0} 2\gamma^{-2} = 0.1560296\dots$ (evaluated in closed form), so $\delta(m) \rightarrow 1$ as $m \rightarrow -\frac{1}{2}^+$. This exhibits the classical density formulation ($m = 0$) and the weighted formulation of Aoki–Koyama ($m = -\frac{1}{2}$) as the two ends of one continuous, quantitatively controlled family — a conditional structural connection between, though not a logical inclusion between, the two definitions compared in Problem 12 of the comparative prime number theory problem list of Hamieh, Kadiri, Martin and Ng, which remains open.

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1 Introduction

Chebyshev’s observation — that primes $\equiv 3 \pmod{4}$ predominate over primes $\equiv 1 \pmod{4}$ for most x — was quantified by Rubinstein and Sarnak [1]: under GRH and the linear independence (LI) of the ordinates of zeros of $L(s, \chi_4)$, the set $\{x : \pi(x; 4, 3) > \pi(x; 4, 1)\}$ has logarithmic density $0.9959\dots$. A separate line of work, initiated by Aoki and Koyama [2] and developed by Sheth [3] and Shimada–Koyama [4], studies *weighted* counts $\sum_{p \leq x, p \equiv a \pmod{q}} p^{-s}$ and locates a transition at $s = \frac{1}{2}$: for the mod-4 race the weighted difference obeys $\sum_{p \leq x} \chi_4(p) p^{-1/2} = -\frac{1}{2} \log \log x + O(1)$ (under DRH in [2], under GRH off a small exceptional set in [3]), so the weighted race is single-signed for large x , while for weights p^{-s} with $s < \frac{1}{2}$ sign changes persist [4]; Hayani [14] has recently obtained the density-one statement at $s = \frac{1}{2}$ under GRH alone, without linear independence. Humphries [5] established the analogous phase structure for weighted sums of the Liouville function, including the limit $\delta_\alpha \rightarrow 1$ as $\alpha \rightarrow \frac{1}{2}^-$. Problem 12 of the comparative prime number theory problem list [6] (posed by Koyama) asks how the weighted formulation relates to the classical logarithmic-density formulation.

Related work. The weighted-race literature just cited concerns constant or decaying weights, and recent activity stays on that side of the boundary. In the absolutely convergent regime (weights $p^{-\sigma}$, $\sigma > 1$; $m = -\sigma < -1$ in our normalization) Yamamuro [19], following Aymone [16], proves unconditional two-sided bounds on the tails $\sum_{n > x} \Lambda(n) \chi(n) n^{-\sigma} / \log n$ through infinitely divisible distributions of Riemann-zeta type — no GRH, no LI, and no logarithmic densities enter. Hayani [20] generalizes the Aoki–Koyama threshold $\frac{1}{2}$ to higher roots in prime *ideal* races, for weights $(N\mathfrak{p})^{-\alpha}$ with $\alpha \geq 0$, under GRH and partly unconditionally — asymptotic and sign statements, not Rubinstein–Sarnak densities, and only nonincreasing weights. For unweighted races, Bailleul, Hayani and Untrau [21] prove a quantitative Kronecker–Weyl theorem in the 1-Wasserstein metric under GRH and an effective linear-independence hypothesis, with rates of convergence to the limiting distributions and conditional bounds for generalized Skewes numbers; an effective, Wasserstein-quantified version of the interpolation theorem below is a natural question we do not pursue. Martin and Ng [23] obtain inclusivity of classical races under GRH plus the existence of sufficiently many self-sufficient zeros, a hypothesis substantially weaker than full LI; it is plausible that the qualitative parts of Theorem 1 (existence of the limiting distribution and $\frac{1}{2} < \delta(m) < 1$) survive a weakening of LI in their direction, but we do not prove this. Finally, Leung [22] proves, under RH and a quantitative form of LI, multivariate Gaussian limit laws with weak negative correlations for smoothed counts of primes in multiple short intervals, with short-interval race applications; the weights there are smoothings in the interval aspect, not powers p^m , and the contact point with this note is the outlook at the end of Section 5. In all of this literature the weight is constant or decaying; we are not aware of prior Rubinstein–Sarnak density computations, or weight-aspect density asymptotics, for growing weights p^m , $m > 0$.

This note treats the two formulations as endpoints of one family — a viewpoint anticipated in the s -parametrized theorems of Shimada–Koyama [4] and in Aymone’s races with a general weight exponent [16]. For $m \in \mathbb{R}$ set

$$D_m(x) = \sum_{\substack{p \leq x \\ p \equiv 3 \pmod{4}}} p^m - \sum_{\substack{p \leq x \\ p \equiv 1 \pmod{4}}} p^m,$$

so $m = 0$ is the classical race and $m = -s$ the Aoki–Koyama weighting; we allow $m > 0$; the

related work reviewed above contains, to our knowledge, no prior treatment of the race with growing weights. Apart from the interpolation theorem of Section 2, our results are computational, at the standard of numerical work in this area (GRH + LI where densities are involved). Every headline number was computed twice: the sieve counts by fully independent implementations, the densities by deterministic inversion against Monte Carlo — two evaluations that share the same zero data and tail model and are independent only in the evaluation step; the shared tail model is bounded rigorously by the certified enclosures of Section 5 (Table 4).

Results

1. Densities across the delicate range. For $m > -\frac{1}{2}$ the limiting logarithmic distribution of the normalized race exists by the framework of Akbary–Ng–Shahabi [7] (cf. Devin [8]) applied to the shifted explicit formula; the normalized race has the almost-periodic form $1 + \sum_{\gamma > 0} 2a_\gamma(m) \cos(\gamma u)$ with

$$a_\gamma(m) = \frac{2m+1}{\sqrt{(m+\frac{1}{2})^2 + \gamma^2}},$$

γ the ordinates of $L(s, \chi_4)$, and the constant 1 normalizing the prime-square bias term. Monte Carlo evaluation under LI (Section 5) gives Table 1. These are, to our knowledge, the first Rubinstein–Sarnak densities computed for a weighted race between primes in arithmetic progressions (for weighted prime sums against main terms, densities exist in the Mertens-race literature [13]; for analytic explicit-formula control of the log-weighted sum $\theta(x)$, cf. Platt–Trudgian [15]).

m	0	1	2	3	8	20	100
$\delta_4(m)$	0.995928	0.797628	0.694520	0.645850	0.571869	0.538455	0.513778
$\delta_3(m)$	0.999063	0.836877	0.723470	0.666552	0.578534	0.540767	—

Table 1: Logarithmic densities of the set where the nonresidue class leads, moduli 4 and 3: deterministic Gil–Pelaez Fourier inversion of the limiting distribution (511 resp. 537 zeros computed to 25 digits, Gaussian tail from the zero-counting density), cross-checked by independent Monte Carlo with $2\text{--}4 \times 10^6$ samples per entry (binomial standard error $\leq 3.6 \times 10^{-4}$; largest deviation 2.2 standard errors over the 14 entries, consistent with binomial sampling), and at $(q, m) = (4, 1)$ by a single 2×10^7 -sample run, 0.797599 ± 0.000090 , consistent with the deterministic entry (0.3 standard errors). The $m = 0$ column reproduces the classical values 0.995928 and 0.999063 (cf. [1, 9]) to all six decimals.

2. Exact computations, $m > 0$. By segmented sieve with exact 128-bit signed accumulators to $X = 10^{10}$: the mod-4 race first favours the residue class at $p = 5$ for every $m \geq 1$ (against Leech’s 26,861 for $m = 0$); lead changes to 10^{10} number 3,082 ($m = 0$), 125,624, 159,942, 199,540 ($m = 1, 2, 3$); the finite- x logarithmic measures at $X = 10^{10}$ — here and throughout, the finite- x logarithmic measure of the leading set is $(\log(X/3))^{-1} \int_3^X \mathbf{1}[D_m(t) > 0] \frac{dt}{t}$ for modulus 4 and $(\log(X/2))^{-1} \int_2^X \mathbf{1}[D_m(t) > 0] \frac{dt}{t}$ for modulus 3, the race being tracked from its first participating prime (3, resp. 2; conventions in Section 5) — are 0.79380, 0.69275, 0.64900 for $m = 1, 2, 3$, within $\lesssim 4 \times 10^{-3}$ of Table 1. For modulus 3 the weighted first flips occur at

$p = 13$ ($m = 1$) and $p = 7$ ($m = 2, 3$), while the classical race, as expected, does not flip below 10^{10} .

3. Dissolution law. Substituting the zero-counting density $\frac{1}{2\pi} \log \frac{qt}{2\pi}$ into the variance of the limiting distribution gives

$$\text{Var}_q(m) \sim 2(m + \tfrac{1}{2}) \log \frac{q(m + \frac{1}{2})}{2\pi}, \quad \delta_q(m) - \tfrac{1}{2} \sim \frac{1}{\sqrt{2\pi \text{Var}_q(m)}} \quad (m \rightarrow \infty),$$

a weight-aspect analogue of the Fiorilli–Martin density asymptotics [9] (cf. Meng [10] for the k -aspect), now a theorem: Section 4 proves it with error term $O(1/(M \log M))$ and a second-order refinement for the density of the limiting variable of any real primitive character under GRH and $L(\frac{1}{2}, \chi) \neq 0$, and under LI(χ_4) for the logarithmic density of the mod-4 race itself (Corollary 2). At $q = 4$, $m = 100$: deterministic 0.513778, Gaussian $\Phi(1/\sigma)$ 0.51379, the law 0.51380. At $q = 3$, $m = 60$: deterministic 0.519716 against the law’s 0.51978 — agreement to 6×10^{-5} . The modulus ordering $\delta_3(m) > \delta_4(m)$ holds at every computed m . In the law it is a conductor effect: $\text{Var}_4(m) - \text{Var}_3(m) \sim 2(m + \frac{1}{2}) \log \frac{4}{3}$, and from the zero data this conductor term accounts for the entire variance gap at $m = 8, 20, 60$ (100–101%). The larger first zero $\gamma_1(\chi_3) = 8.04 > \gamma_1(\chi_4) = 6.02$ pushes in the same direction but is not the mechanism of the law: it contributes 23% of the gap at $m = 8$ and 0.2% at $m = 60$; only at small m ($0 \leq m \leq 3$, where the first zero carries 24–36% of the total variance) is it a comparable driver of the ordering.

4. The boundary $m \leq -\frac{1}{2}$, numerically. Confirming (not discovering) the theorems cited above: with $C^* = \frac{1}{2}(M - \log 2 - \frac{1}{2}) + K_3 - \log L(\frac{1}{2}, \chi_4) = -0.07153$ (M the Meissel–Mertens constant, K_3 the $k \geq 3$ prime-power contribution), the parameter-free prediction $D_{-1/2}(x) = \frac{1}{2} \log \log x + C^* + \varepsilon(x)$ gives 1.4968 ± 0.0172 at $x = 10^{10}$ against the exact value 1.50775; the fitted slope over $[10^6, 10^{10}]$ is 0.494. The fluctuation ε decays like $\sqrt{\Sigma}/\log x$ with $\Sigma = \sum_{\gamma} 2/\gamma^2 = 0.1560296$, visible decade-by-decade in the residuals — a small refinement of the qualitative picture. For $m < -\frac{1}{2}$ the limits are identified with differences of Meissel–Mertens constants for progressions: $\lim D_{-1} = M(4, 3) - M(4, 1) = 0.334981325299993\dots$, our Möbius-over-log L evaluation agreeing with Languasco–Zaccagnini [11] to all compared digits (the difference itself is OEIS A086239; the individual constants are A368645 and A368646). Notably, none of the races $m \in \{-1, -\frac{3}{4}, -\frac{1}{2}\}$ changes sign even once on $[3, 10^{10}]$ (minimum margin $1/\sqrt{3} - 1/\sqrt{5} \approx 0.13014$, attained by the $m = -\frac{1}{2}$ race at $x = 5$; the $m = -1$ margin there is $2/15$): in this range of weights Chebyshev’s bias is not merely predominant but, empirically, total from the first odd prime.

5. An interpolation theorem. Section 2 (Theorem 1) proves: under GRH and LI for $L(s, \chi_4)$, for every $m \in (-\frac{1}{2}, \infty)$ the normalized race has a limiting logarithmic distribution, the density $\delta(m)$ exists and lies in $(\frac{1}{2}, 1)$, the map $m \mapsto \delta(m)$ is continuous, and

$$1 - \delta(m) \leq \min \left\{ C(2m + 1)^2, e^{-1/(2C(2m+1)^2)} \right\}, \quad C = \sum_{\gamma > 0} \frac{2}{\gamma^2} = 0.1560296\dots,$$

so $\delta(m) \rightarrow 1$ as $m \rightarrow -\frac{1}{2}^+$ — a weight-aspect instance of the highly biased phenomenon of Fiorilli [12] — matching the single-sign regime established at $m = -\frac{1}{2}$ by [2, 3, 14]. (The theorem is stated for the log-weighted race; Proposition 1 transfers every statement verbatim to the unlogged race D_m .) The existence statement is an application of the Akbary–Ng–Shahabi

framework [7], and the limit $\delta(m) \rightarrow 1$ is the arithmetic-progression analogue of Humphries's theorem for the weighted Liouville function [5, Thm. 1.5], by a transposed form of his method; what we believe is new is the prime-race statement itself, the continuity of δ in the weight, and the explicit rate with a computable constant.

2 An interpolation theorem

Throughout this section $\chi = \chi_4$ is the nontrivial character mod 4, $\rho = \frac{1}{2} + i\gamma$ runs over the nontrivial zeros of $L(s, \chi)$ (listed with multiplicity), $\text{GRH}(\chi)$ denotes the Riemann hypothesis for $L(s, \chi)$, and $\text{LI}(\chi)$ the linear independence over $\mathbb{Q}$ of the multiset of positive ordinates γ (LI in particular forces all γ to be distinct, i.e. the zeros to be simple; recall also that $L(\frac{1}{2}, \chi_4) = \sum_{k \geq 0} (-1)^k (2k+1)^{-1/2} > 0$ by the alternating series test, so no ordinate vanishes). For $m \in (-\frac{1}{2}, \infty)$ define the (θ -form) weighted race

$$R_m(x) = \sum_{\substack{p \leq x \\ p \equiv 3(4)}} p^m \log p - \sum_{\substack{p \leq x \\ p \equiv 1(4)}} p^m \log p = - \sum_{p \leq x} \chi(p) p^m \log p,$$

and its normalization

$$E_m(x) = (2m+1) x^{-(m+1/2)} R_m(x).$$

Write

$$a_\gamma(m) = \frac{2m+1}{\sqrt{(m+\frac{1}{2})^2 + \gamma^2}}, \quad X_m = 1 + \sum_{\gamma > 0} 2a_\gamma(m) \cos \theta_\gamma,$$

with θ_γ i.i.d. uniform on $[0, 2\pi)$; the series converges a.s. and in L^2 since $\sum_\gamma a_\gamma(m)^2 < \infty$ (the ordinates have density $\ll \log \gamma$). Set

$$C := \sum_{\gamma > 0} \frac{2}{\gamma^2} = 0.1560296498896 \dots$$

Under $\text{GRH}(\chi_4)$ (in force throughout this section) this constant has a closed evaluation: the completed function $\Lambda(s) = (4/\pi)^{(s+1)/2} \Gamma(\frac{s+1}{2}) L(s, \chi_4)$ is entire and satisfies $\Lambda(s) = \Lambda(1-s)$ (the root number of χ_4 is +1), so $\Xi(z) := \Lambda(\frac{1}{2} + z)$ is even; since $\Xi(0) > 0$ (as $L(\frac{1}{2}, \chi_4) > 0$), $\sum_\gamma \gamma^{-2} < \infty$, and, under GRH, the zeros of Ξ are exactly $\pm i\gamma$, the Hadamard factorization takes the genus-zero form $\Xi(z)/\Xi(0) = \prod_{\gamma > 0} (1 + z^2/\gamma^2)$, exactly as for the Riemann ξ -function [17, §12]. Hence

$$C = (\log \Lambda)''(\tfrac{1}{2}) = \tfrac{1}{4} \psi'(\tfrac{3}{4}) + \frac{d^2}{ds^2} \log L(s, \chi_4) \Big|_{s=1/2} = 0.15602964988964488 \dots,$$

with $\psi'(\frac{3}{4}) = \pi^2 - 8G$ (G Catalan's constant), evaluated at 40-digit working precision. Cross-check from the zero data: the first 511 ordinates give 0.152546 and the Riemann–von Mangoldt tail $\int_{\gamma_{511}}^\infty 2t^{-2} dN(t)$ with $dN(t) = (2\pi)^{-1} \log(2t/\pi) dt$ gives 0.003487, total 0.156033, agreeing with the exact value to 3.4×10^{-6} — within the expected $O(10^{-5})$ fluctuation of the zero-counting remainder. (Earlier drafts printed the cruder value 0.156033.)

Theorem 1. *Assume $\text{GRH}(\chi_4)$.*

- (i) For each $m \in (-\frac{1}{2}, \infty)$ the function $u \mapsto E_m(e^u)$ possesses a limiting distribution ν_m : for every bounded continuous $f : \mathbb{R} \rightarrow \mathbb{R}$, $\lim_{Y \rightarrow \infty} \frac{1}{Y} \int_0^Y f(E_m(e^u)) du = \int_{\mathbb{R}} f d\nu_m$. Under $LI(\chi_4)$, ν_m is the law of X_m .
- (ii) Under $LI(\chi_4)$, ν_m is absolutely continuous with support $\mathbb{R}$; the logarithmic density $\delta(m)$ of the set $\{x \geq 2 : R_m(x) > 0\}$ exists, equals $\nu_m((0, \infty))$, satisfies $\frac{1}{2} < \delta(m) < 1$ for every $m \in (-\frac{1}{2}, \infty)$, and $m \mapsto \delta(m)$ is continuous on $(-\frac{1}{2}, \infty)$.
- (iii) Under $LI(\chi_4)$, for all $m \in (-\frac{1}{2}, \infty)$,

$$1 - \delta(m) \leq C(2m + 1)^2 \quad \text{and} \quad 1 - \delta(m) \leq \exp\left(-\frac{1}{2C(2m + 1)^2}\right),$$

with no restriction on m ; the second bound is in fact the stronger of the two for every m . In particular $\delta(m) \rightarrow 1$ as $m \rightarrow -\frac{1}{2}^+$. Both bounds are informative only near the left endpoint: the (stronger) exponential bound drops below the trivial estimate $1 - \delta(m) < \frac{1}{2}$ of part (ii) precisely for $m < m_0 := \frac{1}{2}((2C \log 2)^{-1/2} - 1) = 0.57507\dots$, and the first bound is ≥ 1 for $m \geq \frac{1}{2}(C^{-1/2} - 1) = 0.76580\dots$; for $m \geq m_0$ part (iii) asserts nothing beyond part (ii), and this theorem makes no claim about the rate at which $\delta(m)$ approaches $\frac{1}{2}$ as $m \rightarrow \infty$.

- (iv) Consequently the classical density formulation of Chebyshev's bias at $m = 0$ (logarithmic density $\delta(0) = 0.995928\dots$; 0.9959 in [1], six decimals as in [9] and re-derived here) is joined by the continuous family $\{\delta(m)\}_{m \in (-1/2, \infty)}$, with $\delta(m) \rightarrow 1$ as $m \rightarrow -\frac{1}{2}^+$, to the total-bias regime at the weighted threshold $m = -\frac{1}{2}$, where the unlogged race $D_{-1/2}(x) = \sum_{p \leq x} (-\chi_4(p))p^{-1/2}$ equals $\frac{1}{2} \log \log x + O(1)$ and is eventually of one sign (under DRH [2]; under GRH off a set of finite logarithmic measure [3]; single-signed on a set of logarithmic density one under GRH alone [14]). None of these three endpoint results assumes LI. In the opposite direction the family now covers every positive weight; there the computed densities of Table 1 decrease toward $\frac{1}{2}$, in the manner of the dissolution law of the Results section, which this theorem does not prove.

Remark 1. Part (iii) is an arithmetic-progression analogue of Humphries's theorem that $\delta(P_\alpha) \rightarrow 1$ as $\alpha \rightarrow \frac{1}{2}^-$ for weighted sums of the Liouville function [5, Thm. 1.5]. The mechanism is transposed rather than copied: in [5] the mean of the limiting distribution diverges while the variance stays bounded, whereas here the normalized mean is fixed at 1 while the variance collapses like $C(2m + 1)^2$; in both cases the bias-to-spread ratio diverges, and Humphries already remarks (after his Cor. 7.6) that Chebyshev's inequality suffices for the limit. We claim novelty only for the statement in the prime-race setting, the continuity assertion in (ii), and the explicit constant in (iii). Part (i) does not require LI; it is an application of the framework of Akbary–Ng–Shahabi [7] (cf. Devin [8]).

We isolate the two number-theoretic inputs as lemmas.

Lemma 1 (truncated explicit formula, all truncations). *Let $\chi = \chi_4$ and let $\rho = \beta + i\gamma$ run over the nontrivial zeros of $L(s, \chi)$, listed with multiplicity. Unconditionally, for all $x \geq 2$ and all $T \geq 2$,*

$$\psi(x, \chi) = - \sum_{|\gamma| \leq T} \frac{x^\rho}{\rho} + O\left(\frac{x \log^2(xT)}{T} + \log x\right). \quad (1)$$

No inequality between T and x is assumed; in particular $T > x$ is allowed.

Proof. All implied constants are absolute ($q = 4$ is fixed). We use five standard facts; each is a statement about a single point s or a single height t and contains no parameter x , so no hidden comparison of T with x can enter through them.

(F1) (*Perron kernel*). For $y > 0$, $c > 0$, $T \geq 2$ put $I(y, T) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} y^s s^{-1} ds$ and $\delta(y) = 1, \frac{1}{2}, 0$ according as $y > 1$, $y = 1$, $y < 1$. Then

$$|I(y, T) - \delta(y)| \leq y^c \min\left(1, \frac{1}{\pi T |\log y|}\right) \quad (y \neq 1), \quad \left|I(1, T) - \frac{1}{2}\right| \leq \frac{c}{\pi T}.$$

Proof of (F1). For $y < 1$ close the contour to the right: on the rectangle with right edge at $\operatorname{Re} s = A$ the integrand $y^s s^{-1}$ is analytic, and the right edge is $\ll y^A/A \rightarrow 0$ as $A \rightarrow \infty$; hence I equals minus the two horizontal rays from $c \pm iT$ to $+\infty \pm iT$, each of modulus $\leq \frac{1}{2\pi T} \int_c^\infty y^\sigma d\sigma = \frac{y^c}{2\pi T \log(1/y)}$. For $y > 1$ close to the left, picking up the residue 1 at $s = 0$; the rays to $-\infty \pm iT$ give the same bound with $\log y$. For the $\min(1, \cdot)$ alternative, replace the vertical segment by the arc of the circle $|s| = \sqrt{c^2 + T^2}$ through $c \pm iT$ on the side not containing 0 (for $y > 1$, the side containing 0, picking up the residue 1): on the arc $|y^s| \leq y^c$ and $|s| = \sqrt{c^2 + T^2}$, so the arc integral is $\leq y^c \cdot (\text{length})/(2\pi|s|) \leq y^c$. For $y = 1$ the odd part of the integrand integrates to zero and $I(1, T) = \frac{1}{\pi} \arctan(T/c) = \frac{1}{2} - \frac{1}{\pi} \arctan(c/T)$, with $\arctan(c/T) \leq c/T$. Nothing here involves any second parameter.

(F2) $-\zeta'/\zeta(\sigma) \leq (\sigma - 1)^{-1} + O(1)$ for $1 < \sigma \leq 3$ (simple pole of ζ at 1; [17, §17]).

(F3) $N(t + 1, \chi) - N(t, \chi) \ll \log(t + 2)$ for $t \geq 0$, where $N(t, \chi)$ counts zeros with $0 < \gamma \leq t$; unconditional Riemann–von Mangoldt for $L(s, \chi_4)$ [17, §16]. Zeros are symmetric under $\gamma \mapsto -\gamma$ (χ_4 is real).

(F4) For $-1 \leq \sigma \leq 2$, $s = \sigma + it$, $\frac{L'}{L}(s, \chi) = \sum_{\rho: |\gamma - t| \leq 1} \frac{1}{s - \rho} + O(\log(|t| + 2))$ [17, §16]; and for $2 \leq \sigma \leq 3$, $|L'/L(s, \chi)| \leq \sum_n \Lambda(n) n^{-2} = O(1)$.

(F5) $\Lambda(s) = (4/\pi)^{(s+1)/2} \Gamma(\frac{s+1}{2}) L(s, \chi)$ satisfies $\Lambda(s) = \Lambda(1 - s)$ ([17, §9]; the root number of χ_4 is $\tau(\chi_4)/(i\sqrt{4}) = 2i/2i = +1$). Taking logarithmic derivatives,

$$\frac{L'}{L}(s, \chi) = -\log \frac{4}{\pi} - \frac{1}{2} \psi\left(\frac{s+1}{2}\right) - \frac{1}{2} \psi\left(\frac{2-s}{2}\right) - \frac{L'}{L}(1-s, \chi).$$

If $\sigma \leq -1$ and either $|t| \geq 2$ or σ is a negative even integer, then: $\operatorname{Re}(1 - s) \geq 2$, so $|L'/L(1 - s, \chi)| = O(1)$ by (F4); $\operatorname{Re}(\frac{2-s}{2}) \geq \frac{3}{2}$, so $|\psi(\frac{2-s}{2})| \ll \log(2 + |s|)$ by Stirling; and for $w = \frac{s+1}{2}$ the reflection $\psi(w) = \psi(1 - w) - \pi \cot(\pi w)$ applies with $\operatorname{Re}(1 - w) \geq 1$ and with $|\cot(\pi w)| \leq \coth(\pi)$ when $|\operatorname{Im} w| = |t|/2 \geq 1$, respectively $|\cot(\pi w)| = |\tanh(\pi t/2)| \leq 1$ when $\operatorname{Re} w$ is a half-odd-integer (which is the case when σ is a negative even integer). Hence on those sets $|L'/L(s, \chi)| \ll \log(2 + |s|)$.

Also $L(0, \chi_4) = \frac{1}{2} \neq 0$: evaluating $\Lambda(0) = \Lambda(1)$ from (F5), $(4/\pi)^{1/2} \Gamma(\frac{1}{2}) L(0, \chi_4) = (4/\pi) \Gamma(1) L(1, \chi_4)$, and $L(1, \chi_4) = \pi/4$ (Leibniz), so $L(0, \chi_4) = (4/\pi)^{1/2} (\pi/4) / \sqrt{\pi} = \frac{1}{2}$. Consequently L'/L is regular at $s = 0$. Moreover every nontrivial zero has $0 < \beta < 1$: there are no zeros with $\beta = 1$ (de la Vallée Poussin, unconditional), hence none with $\beta = 0$ by (F5); so the pole of $1/s$ at $s = 0$ collides with no zero.

Step 1 (Perron). Let $c = 1 + 1/\log x \in (1, 1 + 1/\log 2]$, so $x^c = ex$. On $\operatorname{Re} s = c$ the Dirichlet series $-L'/L(s, \chi) = \sum_n \Lambda(n) \chi(n) n^{-s}$ converges absolutely, so term-by-term application of

(F1) on the finite segment gives

$$\psi(x, \chi) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \left(-\frac{L'}{L}(s, \chi) \right) \frac{x^s}{s} ds + O(R_1), \quad R_1 = \sum_{n \geq 2} \Lambda(n) \left(\frac{x}{n} \right)^c \min \left(1, \frac{1}{T |\log(x/n)|} \right) + \log x,$$

the final $\log x$ covering the possible term $n = x$ (where (F1) gives $\frac{1}{2}$ instead of the weight 1 used in ψ , a discrepancy $\leq \Lambda(x) \ll \log x$). Split R_1 : for $n \leq \frac{3}{4}x$ or $n \geq \frac{5}{4}x$ one has $|\log(x/n)| \geq \log \frac{5}{4}$, so these terms contribute $\ll T^{-1} x^c \sum_n \Lambda(n) n^{-c} = T^{-1} x^c (-\zeta'/\zeta(c)) \ll (x/T) \log x$ by (F2). For $\frac{3}{4}x < n < \frac{5}{4}x$, $n \neq x$: here $\Lambda(n) \ll \log x$, $(x/n)^c \ll 1$, and $|\log(x/n)| \geq |n-x|/(2x)$, so these terms contribute

$$\ll \log x \sum_{1 \leq k \leq x} \min \left(1, \frac{2x}{Tk} \right) \ll \log x \left(1 + \frac{x}{T} + \frac{x}{T} \log x \right) \ll \log x + \frac{x \log^2 x}{T},$$

where k runs over the possible values of $\lceil |n-x| \rceil$ (the at most two integers nearest x are covered by the $\min(\cdot) \leq 1$ bound). Hence $R_1 \ll x \log^2 x / T + \log x$, for every $T \geq 2$; when $T > x$ these bounds only become smaller.

Step 2 (choice of height T'). Let $G = \{\gamma : T-1 < \gamma < T+2\}$; by (F3), $\#G \leq A_1 \log(T+2)$ for an absolute A_1 . With $r = 1/(4A_1 \log(T+2))$ the union of the intervals $(\gamma-r, \gamma+r)$, $\gamma \in G$, has measure $\leq \frac{1}{2}$, so there exists $T' \in [T, T+1]$ with $|T' - \gamma| \geq r$ for all $\gamma \in G$; for $\gamma \notin G$, $|T' - \gamma| \geq 1$ automatically. Thus $|T' - \gamma| \gg 1/\log(T+2)$ for every ordinate γ , and by the $\gamma \mapsto -\gamma$ symmetry the same holds at height $-T'$.

Step 3 (contour). For $K \in \mathbb{N}$ set $U = 2K$ and integrate $(-L'/L)(s, \chi) x^s/s$ counterclockwise around the rectangle with vertices $c \pm iT'$, $-U \pm iT'$. The poles inside are: the nontrivial zeros with $|\gamma| \leq T'$ (residue $-x^\rho/\rho$ each, with multiplicity; $\rho \neq 0$ since $0 < \beta < 1$); the simple pole of $1/s$ at $s = 0$ (residue $-L'/L(0, \chi) = O(1)$, using $L(0, \chi) = \frac{1}{2}$); and the trivial zeros $s = -(2j+1)$, $0 \leq j \leq K-1$, of $L(s, \chi_4)$ (simple, from the poles of $\Gamma(\frac{s+1}{2})$; residue $+x^{-(2j+1)}/(2j+1)$ each). Hence

$$\frac{1}{2\pi i} \oint = - \sum_{|\gamma| \leq T'} \frac{x^\rho}{\rho} - \frac{L'}{L}(0, \chi) + \sum_{j=0}^{K-1} \frac{x^{-(2j+1)}}{2j+1}.$$

Step 4 (horizontal segments, $-1 \leq \sigma \leq c$). On $s = \sigma \pm iT'$ with $-1 \leq \sigma \leq 2$, (F4) and Step 2 give $|L'/L(s, \chi)| \leq \sum_{|\gamma \mp T'| \leq 1} |T' - \gamma|^{-1} + O(\log T') \ll \log^2(T+2)$ ($\ll \log(T+2)$ terms, each $\ll \log(T+2)$); for $2 \leq \sigma \leq c$ (nonempty only if $x < e$) the $O(1)$ bound of (F4) applies. Hence these pieces contribute

$$\ll \frac{\log^2(T+2)}{T} \int_{-1}^c x^\sigma d\sigma \ll \frac{\log^2(T+2)}{T} \cdot \frac{x^c}{\log x} \ll \frac{x \log^2(xT)}{T}.$$

Step 5 (horizontal segments $-U \leq \sigma \leq -1$, and vertical segment $\sigma = -U$). By (F5), $|L'/L| \ll \log(2+|s|)$ on both. The horizontal parts contribute

$$\ll \frac{1}{T} \int_1^\infty x^{-v} \log(3+v+T) dv \ll \frac{1}{T} \left(\frac{\log(3+T)}{x \log x} + \frac{1}{x} \right) \ll \frac{\log(3+T)}{T} \ll 1,$$

uniformly in U , absorbed in $O(\log x)$. The vertical segment has length $2T'$, $|x^s/s| \leq x^{-U}/U$, so it contributes $\ll T' x^{-U} \log(U+T'+2)/U \rightarrow 0$ as $K \rightarrow \infty$ (here $x \geq 2 > 1$).

Step 6 (assemble; $U \rightarrow \infty$). Letting $K \rightarrow \infty$, the trivial-zero sum increases to $\sum_{j \geq 0} x^{-(2j+1)}/(2j+1) = \operatorname{artanh}(1/x) \leq \operatorname{artanh}(\frac{1}{2}) < 0.55$. Combining Steps 1–5,

$$\psi(x, \chi) = - \sum_{|\gamma| \leq T'} \frac{x^\rho}{\rho} + O\left(\frac{x \log^2(xT)}{T} + \log x\right).$$

Step 7 ($T' \rightarrow T$). The zeros with $T < |\gamma| \leq T' \leq T + 1$ number $\ll \log(T + 2)$ by (F3), and each satisfies $|x^\rho/\rho| \leq x^\beta/|\gamma| \leq x/T$; restoring the summation range to $|\gamma| \leq T$ therefore costs $\ll x \log(T + 2)/T \ll x \log^2(xT)/T$. This proves (1). Every estimate above is uniform in $T \geq 2$, and no step compared T with x . $\square$

Lemma 2 (shifted truncated explicit formula). *Assume $\operatorname{GRH}(\chi_4)$ and let $m \in (-\frac{1}{2}, \infty)$ be fixed. For $\psi_m(x) := \sum_{n \leq x} \Lambda(n) \chi(n) n^m$ and all $x \geq 2$, $T \geq 2$ (no inequality between T and x is assumed; Lemma 1),*

$$\psi_m(x) = - \sum_{|\gamma| \leq T} \frac{x^{\rho+m}}{\rho+m} + O_m\left(\frac{x^{m+1} \log^2(xT)}{T} + (1+x^m) \log x\right). \quad (2)$$

For $m \leq 0$ the floor $(1+x^m) \log x \leq 2 \log x$ recovers the earlier form of this lemma.

Proof. Since $\psi(t, \chi) = 0$ for $t < 3$, Riemann–Stieltjes partial summation gives exactly

$$\psi_m(x) = \int_{2-}^x t^m d\psi(t, \chi) = x^m \psi(x, \chi) - m \int_2^x t^{m-1} \psi(t, \chi) dt.$$

Insert (1) (Lemma 1, valid for every $T \geq 2$), with the same T , at x and at every $t \in [2, x]$. The zero terms produce, for each ρ with $|\gamma| \leq T$,

$$x^m \frac{x^\rho}{\rho} - m \int_2^x t^{m-1} \frac{t^\rho}{\rho} dt = \frac{x^{\rho+m}}{\rho+m} + \frac{m 2^{\rho+m}}{\rho(\rho+m)},$$

an identity (integrate $t^{\rho+m-1}$; note $\operatorname{Re}(\rho+m) = m + \frac{1}{2} > 0$ for every $m > -\frac{1}{2}$, and $|\rho+m| \geq \gamma \geq \gamma_1 > 6$, so no denominator degenerates at any m in the range). The endpoint terms are *summable*: since $|\rho| \geq \gamma$ and $|\rho+m| \geq \gamma$,

$$\sum_{|\gamma| \leq T} \left| \frac{m 2^{\rho+m}}{\rho(\rho+m)} \right| \leq |m| 2^{m+1/2} \sum_{|\gamma| \leq T} \frac{1}{\gamma^2} \ll_m 1,$$

uniformly in T . (Bounding each term by $O_\gamma(1)$ alone would not suffice, as there are $\asymp T \log T$ of them.) The error terms of (1) contribute

$$x^m \cdot O\left(\frac{x \log^2(xT)}{T} + \log x\right) + |m| \int_2^x t^{m-1} O\left(\frac{t \log^2(tT)}{T} + \log t\right) dt.$$

The truncation parts are $\ll_m x^{m+1} \log^2(xT)/T$, using $\log^2(tT) \leq \log^2(xT)$ and $\int_2^x t^m dt \leq x^{m+1}/(m+1)$ for $m > -1$. The log-floor parts split into three cases:

$$|m| \int_2^x t^{m-1} \log t dt \ll_m \begin{cases} x^m \log x, & m > 0 \quad (\leq \log x \cdot \int_2^x m t^{m-1} dt \leq x^m \log x), \\ 0, & m = 0 \text{ (the factor } m \text{ vanishes)}, \\ 1, & m < 0 \quad (\int_2^\infty t^{m-1} \log t dt = O_m(1)). \end{cases}$$

Since $\log x \geq \log 2$, every $O_m(1)$ above (including the endpoint sum) is $\ll_m \log x$, so all non-zero-sum contributions are $O_m(x^{m+1} \log^2(xT)/T + (1+x^m) \log x)$, proving (2). Neither half of the floor can be dropped on the full range: for $m < 0$ the quantity $x^m \log x$ tends to 0 while genuinely $O_m(1)$ terms (the endpoint sum, the constants inside (1)) persist, so the summand 1 is needed; for $m > 0$ the term $x^m \log x$ arises from $m \int_2^x t^{m-1} O(\log t) dt$, and removing it would require sign cancellation in the $O(\log t)$ errors of (1) that we do not know. After the normalization by $(2m+1)x^{-(m+1/2)}$, the floor becomes, with $x = e^u$,

$$(1 + e^{mu}) u e^{-(m+1/2)u} = u(e^{-(m+1/2)u} + e^{-u/2}) \leq 2u e^{-\kappa_m u}, \quad \kappa_m := \min(m + \tfrac{1}{2}, \tfrac{1}{2}) > 0,$$

an $L^2(0, \infty)$ function of u for every $m > -\frac{1}{2}$; the mean-square hypothesis (3) used in the proof of (i) is therefore unaffected by the extension. $\square$

Lemma 3 (prime-power separation). *Fix $m \in (-\frac{1}{2}, \infty)$. For $x \geq 4$,*

$$-R_m(x) = \psi_m(x) - \sum_{p \leq \sqrt{x}} p^{2m} \log p + O_m(x^{m+1/3} \log x + \log^2 x),$$

and

$$\sum_{p \leq \sqrt{x}} p^{2m} \log p = \frac{x^{m+1/2}}{2m+1} + O_m(x^{m+1/2} e^{-c\sqrt{\log x}})$$

with $c > 0$ absolute and the implied constants depending on m . Both error terms are $o(x^{m+1/2})$. The $\log^2 x$ term originates solely in the prime powers p^k , $k \geq 3$, with $km \leq -1$; such k exist only for $m < 0$, and for $m > -\frac{1}{3}$ the term is absorbed by $O_m(x^{m+1/3} \log x)$, so it is genuinely present only for $m \in (-\frac{1}{2}, -\frac{1}{3}]$.

Proof. ψ_m splits over prime powers $p^k \leq x$. The $k = 1$ part is $\sum_{p \leq x} \chi(p) p^m \log p = -R_m(x)$. The $k = 2$ part is $\sum_{p^2 \leq x} \chi(p)^2 p^{2m} \log p = \sum_{3 \leq p \leq \sqrt{x}} p^{2m} \log p = \sum_{p \leq \sqrt{x}} p^{2m} \log p + O_m(1)$, since $\chi(p)^2 = 1$ for odd p and $\chi(2) = 0$.

The $k \geq 3$ terms. Here $3 \leq k \leq \log x / \log 2$ and $y := x^{1/k}$. Let A be Chebyshev's absolute constant in $\theta(y) \leq Ay$.

If $m \geq 0$ then $km \geq 0 > -1$ for every k — no case split occurs — and monotonicity gives, uniformly in k ,

$$\sum_{p \leq y} p^{km} \log p \leq y^{km} \theta(y) \leq Ay^{km+1} = Ax^{(km+1)/k} = Ax^{m+1/k} \leq Ax^{m+1/3};$$

summing over the $\leq \log x / \log 2$ values of k yields $O(x^{m+1/3} \log x)$ with an absolute constant.

If $m < 0$, split on the sign of $km + 1$. For the (fewer than $1/|m|$, hence $O_m(1)$ many) k with $-1 < km < 0$, partial summation with $\theta(t) \leq At$ gives

$$\sum_{p \leq y} p^{km} \log p = y^{km} \theta(y) + |km| \int_2^y t^{km-1} \theta(t) dt \leq Ay^{km+1} \left(1 + \frac{|km|}{km+1}\right) \ll_m x^{m+1/k} \leq x^{m+1/3},$$

the constant depending on m through the least positive value of $km + 1$ over the finitely many admissible k ; the total over these k is $O_m(x^{m+1/3})$. For the k with $km \leq -1$ (these exist iff $m < 0$, namely $k \geq 1/|m|$), we have $p^{km} \log p \leq p^{-1} \log p$, so by Mertens

$$\sum_{p \leq y} p^{km} \log p \leq \sum_{p \leq y} \frac{\log p}{p} = \log y + O(1) \ll \log x,$$

uniformly in k ; over $\leq \log x / \log 2$ values of k the total is $O(\log^2 x)$. Combining the cases, the $k \geq 3$ contribution is $O_m(x^{m+1/3} \log x + \log^2 x)$ for every $m \in (-\frac{1}{2}, \infty)$, and the provenance stated in the lemma is exactly the case analysis just made; for $m > -\frac{1}{3}$ one has $\log x \ll_m x^{m+1/3}$, so there the $\log^2 x$ term may be dropped — we keep it so that a single statement covers the whole range. Both error terms are $o(x^{m+1/2})$ uniformly: $x^{m+1/3} \log x / x^{m+1/2} = x^{-1/6} \log x \rightarrow 0$ and $\log^2 x \cdot x^{-(m+1/2)} \rightarrow 0$ since $m + \frac{1}{2} > 0$. (The crude bound “ $O(x^{1/3})$ prime powers times $\log x$ ” is *not* sufficient here: $x^{1/3} \geq x^{m+1/2}$ for $m \leq -\frac{1}{6}$.)

The squares sum. The argument is unchanged from the $(-\frac{1}{2}, 0]$ version, and we record that its only structural input is $2m + 1 > 0$, so it holds on the whole range. Write $\theta(y) = y + E(y)$, $E(y) \ll ye^{-2c\sqrt{\log y}}$ (prime number theorem with the classical de la Vallée Poussin error). Partial summation gives, with $y = \sqrt{x}$,

$$\sum_{p \leq y} p^{2m} \log p = y^{2m} \theta(y) - 2m \int_2^y t^{2m-1} \theta(t) dt = \frac{y^{2m+1}}{2m+1} + O_m(1) + y^{2m} E(y) - 2m \int_2^y t^{2m-1} E(t) dt,$$

the $O_m(1)$ being the endpoint term $2m 2^{2m+1} / (2m+1)$. Here $y^{2m} E(y) \ll y^{2m+1} e^{-2c\sqrt{\log y}}$, and splitting the integral at $\sqrt{y}$,

$$\int_2^y t^{2m} e^{-2c\sqrt{\log t}} dt \leq \frac{y^{(2m+1)/2}}{2m+1} + e^{-2c\sqrt{(\log y)/2}} \frac{y^{2m+1}}{2m+1} \ll_m y^{2m+1} e^{-c\sqrt{\log y}},$$

where both steps use only $2m + 1 > 0$: the first integral was extended using $\int_2^{\sqrt{y}} t^{2m} dt \leq y^{(2m+1)/2} / (2m+1)$, and $y^{(2m+1)/2} \ll_m y^{2m+1} e^{-c\sqrt{\log y}}$ because $e^{c\sqrt{\log y}}$ grows slower than any positive power of y . As $y^{2m+1} = x^{m+1/2} \rightarrow \infty$, the $O_m(1)$ is absorbed. All constants depend on m : they blow up like $(2m+1)^{-1}$ as $m \rightarrow -\frac{1}{2}^+$ and (through 2^{2m+1} and the factor $2m+1$) grow with m as $m \rightarrow \infty$; the theorem is for fixed m , and no uniformity in m is used anywhere below — the continuity in (ii) is proved at the level of the limit random variables, not of the explicit formula. $\square$

Proof of (i). We verify the hypotheses of [7, Thm. 1.2 / Cor. 1.3(a)] literally. Their setting: $\phi(y) = c + \operatorname{Re} \sum_{\lambda_n \leq X} r_n e^{i\lambda_n y} + E(y, X)$ for all $y \geq y_0$ and all $X \geq X_0$ (*their* (1.7)), with

$$\lim_{Y \rightarrow \infty} \frac{1}{Y} \int_{y_0}^Y |E(y, e^Y)|^2 dy = 0 \quad (3)$$

(*their* (1.8); note the truncation point is e^Y , tied to the averaging window — this is why Lemma 2 must allow $T > x$). Corollary 1.3(a) then requires $r_n \ll \lambda_n^{-\beta}$ for some $\beta > \frac{1}{2}$ and the counting bound $\sum_{T < \lambda_n \leq T+1} 1 \ll \log T$.

Take $\phi(u) = E_m(e^u)$, $y_0 = 2$, $c = 1$, $\lambda_n = \gamma_n$ the positive ordinates in increasing order (with multiplicity), and

$$r_\gamma(m) = \frac{2(2m+1)}{m + \frac{1}{2} + i\gamma}, \quad \text{so } |r_\gamma(m)| = 2a_\gamma(m).$$

Pairing ρ and $\bar{\rho}$ in Lemma 2 (the zeros of $L(s, \chi_4)$ are symmetric under conjugation since χ_4 is real) and combining with Lemma 3, for $u \geq 2$ and any $T \geq 2$,

$$E_m(e^u) = 1 + \operatorname{Re} \sum_{0 < \gamma \leq T} r_\gamma(m) e^{i\gamma u} + \mathcal{E}(u, T),$$

$$\mathcal{E}(u, T) \ll_m \frac{e^{u/2} \log^2(e^u T)}{T} + u^2 e^{-\kappa_m u} + u e^{-u/6} + e^{-c\sqrt{u}}, \quad \kappa_m := \min(m + \tfrac{1}{2}, \tfrac{1}{2}) > 0.$$

(The four terms come from, in order: the truncation error of (2) times $(2m+1)x^{-(m+1/2)}$; the $(1+x^m)\log x$ floor of (2) together with the $\log^2 x$ prime-power floor of Lemma 3, which after normalization are $(1+e^{mu})u e^{-(m+1/2)u} + u^2 e^{-(m+1/2)u} \leq 3u^2 e^{-\kappa_m u}$, since $e^{-(m+1/2)u} \leq e^{-\kappa_m u}$ and $e^{mu} e^{-(m+1/2)u} = e^{-u/2} \leq e^{-\kappa_m u}$; the $k \geq 3$ term $x^{m+1/3} \log x \cdot x^{-(m+1/2)} = x^{-1/6} \log x = u e^{-u/6}$; and the PNT error. All exponents are strictly negative for every $m > -\frac{1}{2}$, which is the only property used.) This establishes their (1.7) with our $\mathcal{E}$ defined by the display, for every $T \geq 2$; square-integrability of ϕ on $[0, y_0]$ is trivial since E_m is bounded there.

For (3), put $T = e^Y$ and $u \in [2, Y]$, so $e^u \leq e^Y = T$: the first term is $\ll e^{u/2} (2Y)^2 e^{-Y}$, whose square integrates to $\frac{1}{Y} \int_2^Y 16Y^4 e^{u-2Y} du \ll Y^3 e^{-Y} \rightarrow 0$; each of the remaining terms is a fixed $L^2(0, \infty)$ function of u (here $\kappa_m > 0$ is used), so its contribution to the Cesàro mean square is $O_m(1/Y) \rightarrow 0$.

The coefficient conditions: $|r_\gamma(m)| \leq 2(2m+1)/\gamma$, so $\beta = 1 > \frac{1}{2}$; and $\sum_{T < \gamma \leq T+1} 1 \ll \log T$ is the Riemann–von Mangoldt formula for $L(s, \chi_4)$, $N(T, \chi_4) = \frac{T}{2\pi} \log \frac{2T}{\pi e} + O(\log T)$ ([17, §16]), which holds unconditionally. Hence [7, Cor. 1.3(a)] applies verbatim and ν_m exists. (The only structural difference from their unshifted applications is that $\text{Re}(\rho + m) = m + \frac{1}{2}$ replaces $\frac{1}{2}$; it is constant along the zeros, which is all their hypotheses see. Shifted denominators $\rho - \alpha$ of exactly this type already occur in their weighted Möbius and Liouville applications [7, Lem. 4.3, Cor. 1.6].)

Under $\text{LI}(\chi_4)$ the multiset $\{\gamma\}$ is linearly independent over $\mathbb{Q}$, and [7, Thm. 1.9] (proved in their §5) gives

$$\hat{\nu}_m(\xi) = e^{-i\xi} \prod_{\gamma > 0} J_0(2a_\gamma(m)\xi),$$

which is precisely the characteristic function (in their Fourier normalization) of $X_m = 1 + \sum 2a_\gamma \cos \theta_\gamma$; by uniqueness of Fourier transforms of probability measures, ν_m is the law of X_m . $\square$

Proof of (ii). Fix $m \in (-\frac{1}{2}, \infty)$ and write $V_m = X_m - 1 = \sum_{\gamma > 0} 2a_\gamma \cos \theta_\gamma$ (convergent a.s. and in L^2).

Absolute continuity. Each summand $2a_\gamma \cos \theta_\gamma$ has the arcsine law with density $\pi^{-1}(4a_\gamma^2 - x^2)^{-1/2}$ on $(-2a_\gamma, 2a_\gamma)$, which is absolutely continuous ($a_\gamma > 0$ because $2m+1 > 0$). Split off the first zero: $V_m = 2a_{\gamma_1} \cos \theta_{\gamma_1} + W$ with W independent of the first summand. If $\mu \ll \text{Leb}$ and ν is any probability law, then for Lebesgue-null A , $(\mu * \nu)(A) = \int \mu(A - w) \nu(dw) = 0$; hence $\text{law}(V_m) = \text{arcsine} * \text{law}(W)$ is absolutely continuous. One absolutely continuous independent component suffices; in particular $\nu_m(\{x\}) = 0$ for every x .

Existence of the density $\delta(m)$. Since $\nu_m(\partial(0, \infty)) = \nu_m(\{0\}) = 0$, sandwiching $\mathbf{1}_{(0, \infty)}$ between bounded continuous functions in the defining property of ν_m shows that the logarithmic density of $\{x : E_m(x) > 0\} = \{x : R_m(x) > 0\}$ exists and equals $\nu_m((0, \infty))$; this is the same passage as [7, eq. (1.22)] and [1, §2].

Full support. We show that $\mathbb{P}(|V_m - v| < \varepsilon) > 0$ for every $v \in \mathbb{R}$ and $\varepsilon > 0$; this gives $\text{supp}(V_m) = \mathbb{R}$. Let $\gamma_1 < \gamma_2 < \dots$ be the positive ordinates (distinct under LI) and $c_n := 2a_{\gamma_n}(m) > 0$. Three facts: (α) $c_n \rightarrow 0$, since $a_\gamma(m) \leq (2m+1)/\gamma$; (β) $\sum_n c_n = \infty$: for every $\gamma \geq m + \frac{1}{2}$ — all but finitely many ordinates — one has $(m + \frac{1}{2})^2 \leq \gamma^2$, hence $a_\gamma(m) \geq (2m+1)/(\gamma\sqrt{2})$, and $\sum_{\gamma \leq T} \gamma^{-1} \gg (\log T)^2$ by partial summation against $N(T, \chi_4) \sim$

$\frac{T}{2\pi} \log \frac{2T}{\pi e}; (\gamma) \sum_n c_n^2 < \infty$. By symmetry of V_m we may take $v \geq 0$ (for $v < 0$ replace $\cos \theta \approx 1$ by $\cos \theta \approx -1$ in Step 3).

Step 1 (a finite block reaching v to within $\varepsilon/4$). By (α) fix N_0 with $c_n < \varepsilon/4$ for all $n > N_0$. By (β) , $\sum_{n > N_0} c_n = \infty$, so there is a minimal $N_1 \geq N_0$ with $S := \sum_{n=N_0+1}^{N_1} c_n \geq v$ (if $v = 0$ take $N_1 = N_0$, $S = 0$). Minimality and $c_{N_1} < \varepsilon/4$ give $v \leq S < v + \varepsilon/4$. Put $B = \{N_0 + 1, \dots, N_1\}$.

Step 2 (tail). By (γ) fix $N \geq N_1$ with $\sum_{n > N} 2a_{\gamma_n}^2 \leq \varepsilon^2/64$. The tail $W := \sum_{n > N} 2a_{\gamma_n} \cos \theta_{\gamma_n}$ (convergent a.s. and in L^2) has mean 0 and $\text{Var}(W) = \sum_{n > N} 2a_{\gamma_n}^2 \leq \varepsilon^2/64$ (each summand has variance $2a_{\gamma_n}^2$), so Chebyshev gives $\mathbb{P}(|W| \geq \varepsilon/4) \leq (\varepsilon^2/64)(16/\varepsilon^2) = \frac{1}{4}$, i.e. $\mathbb{P}(|W| < \varepsilon/4) \geq \frac{3}{4}$.

Step 3 (head). Put $\delta := \varepsilon/(4(1 + \sum_{n \leq N} c_n)) > 0$ and

$$A := \bigcap_{n \in B} \{|\cos \theta_{\gamma_n} - 1| < \delta\} \cap \bigcap_{\substack{n \leq N \\ n \notin B}} \{|\cos \theta_{\gamma_n}| < \delta\}.$$

Each event has positive probability (θ_{γ_n} is uniform on $[0, 2\pi)$, and $\cos \theta$ lies within δ of 1, resp. of 0, on a set of positive measure), the events concern distinct phases, and the phases are independent, so $\mathbb{P}(A) > 0$. On A the head $H := \sum_{n \leq N} c_n \cos \theta_{\gamma_n}$ satisfies $|H - S| \leq \delta \sum_{n \leq N} c_n < \varepsilon/4$.

Step 4 (combine). A and $\{|W| < \varepsilon/4\}$ involve disjoint sets of independent phases, hence are independent, and $V_m = H + W$ a.s. On their intersection, $|V_m - v| \leq |H - S| + |S - v| + |W| < \frac{3}{4}\varepsilon < \varepsilon$, and

$$\mathbb{P}(|V_m - v| < \varepsilon) \geq \frac{3}{4} \mathbb{P}(A) > 0.$$

Every open interval therefore carries positive mass: $\text{supp}(V_m) = \mathbb{R}$ (cf. the positivity-of-densities argument of [1], of which this is the quantitative form).

Strict inequalities. V_m is symmetric and atomless, so $\mathbb{P}(V_m \geq 0) = \frac{1}{2}$ and $\delta(m) = \mathbb{P}(V_m > -1) = \frac{1}{2} + \mathbb{P}(-1 < V_m < 0) > \frac{1}{2}$, while $1 - \delta(m) = \mathbb{P}(V_m \leq -1) > 0$; both strict by full support. Hence $\frac{1}{2} < \delta(m) < 1$.

Continuity. The characteristic function of V_m is $\Phi_m(t) = \prod_{\gamma} J_0(2a_{\gamma}(m)t)$ (dominated convergence from the finite products). Fix $t_0 \geq 1$ and a compact $K \subset (-\frac{1}{2}, \infty)$, and put

$$A_K := 2 \max K + 1 \in (0, \infty), \quad \text{so} \quad a_{\gamma}(m') \leq \frac{2m' + 1}{\gamma} \leq \frac{A_K}{\gamma} \quad \text{for all } m' \in K, \gamma > 0.$$

For $\Gamma \geq 2A_K t_0$ and $\gamma > \Gamma$ we have $|2a_{\gamma}(m')t| \leq 2A_K t_0/\gamma \leq 1$, and for $|z| \leq 1$ the alternating series gives $J_0(z) \geq 1 - z^2/4 \geq \frac{3}{4}$, whence $|\log J_0(z)| \leq z^2/3$. Therefore

$$\left| \sum_{\gamma > \Gamma} \log J_0(2a_{\gamma}(m')t) \right| \leq \frac{4A_K^2 t_0^2}{3} \sum_{\gamma > \Gamma} \gamma^{-2} =: \eta_K(\Gamma) \rightarrow 0 \quad (\Gamma \rightarrow \infty),$$

uniformly in $m' \in K$ and $|t| \leq t_0$. Writing $\Phi_{m'}$ as (finite product over $\gamma \leq \Gamma$, jointly continuous in (m', t) since each $a_{\gamma}(\cdot)$ is continuous on $(-\frac{1}{2}, \infty) \times \exp(\sum_{\gamma > \Gamma} \log J_0)$, and noting all factors have modulus ≤ 1 , a standard 3ε argument gives $\Phi_{m'}(t) \rightarrow \Phi_m(t)$ for every t as $m' \rightarrow m$ within $(-\frac{1}{2}, \infty)$. Each $\Phi_{m'}$ is a characteristic function and the limit Φ_m is continuous at 0, so by Lévy's continuity theorem $X_{m'} \Rightarrow X_m$ weakly. Since $\nu_m(\{0\}) = 0$, the portmanteau theorem for the set $(0, \infty)$ gives $\delta(m') \rightarrow \delta(m)$: the map $m \mapsto \delta(m)$ is continuous on $(-\frac{1}{2}, \infty)$. $\square$

Proof of (iii). Let $V_m = X_m - 1$ as above; $\mathbb{E}V_m = 0$ and, by independence and L^2 convergence,

$$\sigma_m^2 := \text{Var}(V_m) = \sum_{\gamma>0} 2a_\gamma(m)^2 \leq (2m+1)^2 \sum_{\gamma>0} \frac{2}{\gamma^2} = C(2m+1)^2,$$

since $a_\gamma(m)^2 \leq (2m+1)^2/\gamma^2$. Chebyshev's inequality gives

$$1 - \delta(m) = \mathbb{P}(V_m \leq -1) \leq \mathbb{P}(|V_m| \geq 1) \leq \sigma_m^2 \leq C(2m+1)^2.$$

(Cantelli's one-sided inequality would give the marginally better $\sigma_m^2/(1 + \sigma_m^2)$; we keep the clean form.)

For the exponential bound, set $A := \sum_{\gamma>0} a_\gamma(m)^2 = \sigma_m^2/2 \leq \frac{C}{2}(2m+1)^2$ and let $S_N = \sum_{n \leq N} 2a_{\gamma_n} \cos \theta_{\gamma_n}$. For $s > 0$, independence gives

$$\mathbb{E} e^{-sS_N} = \prod_{n \leq N} I_0(2a_{\gamma_n}s) \leq \exp\left(s^2 \sum_{n \leq N} a_{\gamma_n}^2\right) \leq e^{s^2 A},$$

where I_0 is the modified Bessel function, $\mathbb{E} e^{x \cos \theta} = I_0(x)$, and $I_0(x) = \sum_k (x/2)^{2k}/(k!)^2 \leq \sum_k (x^2/4)^k/k! = e^{x^2/4}$ term by term (Montgomery's inequality; it is inequality (33) of [5], from [18, §3, eq. (2)]). Chernoff's bound with the optimal $s = (1 - \varepsilon)/(2A)$ yields, for $0 < \varepsilon < 1$,

$$\mathbb{P}(S_N \leq -(1 - \varepsilon)) \leq \exp\left(-\frac{(1 - \varepsilon)^2}{4A}\right).$$

Since $S_N \rightarrow V_m$ in L^2 , $\mathbb{P}(V_m \leq -1) \leq \mathbb{P}(S_N \leq -(1 - \varepsilon)) + \mathbb{P}(|V_m - S_N| \geq \varepsilon)$, and letting $N \rightarrow \infty$ then $\varepsilon \rightarrow 0^+$,

$$1 - \delta(m) = \mathbb{P}(V_m \leq -1) \leq \exp\left(-\frac{1}{4A}\right) \leq \exp\left(-\frac{1}{2C(2m+1)^2}\right).$$

Both stated bounds follow, for every $m \in (-\frac{1}{2}, \infty)$ — no step above used $m \leq 0$ — and $e^{-1/(2z)} < z$ for all $z > 0$ (equivalent to $\log w < w/2$ at $w = 1/z$), so the exponential bound is always the stronger one. $\delta(m) \rightarrow 1$ as $m \rightarrow -\frac{1}{2}^+$ is immediate.

Scope. Although valid on the whole range, the bounds carry information only near the left endpoint, and we state this openly. With $C = 0.1560296\dots$: the Chebyshev bound $C(2m+1)^2$ is below the trivial estimate $\frac{1}{2}$ of part (ii) only for $m < \frac{1}{2}((2C)^{-1/2} - 1) = 0.39506\dots$ and is ≥ 1 for $m \geq \frac{1}{2}(C^{-1/2} - 1) = 0.76580\dots$; the exponential bound is below $\frac{1}{2}$ only for $m < m_0 = \frac{1}{2}((2C \log 2)^{-1/2} - 1) = 0.57507\dots$. For $m \geq m_0$, part (iii) asserts nothing beyond part (ii). This limitation is consistent with the expected truth: the computed densities approach $\frac{1}{2}$ from above (Table 1), so an upper bound on $1 - \delta(m)$ tending to 0 for large m would contradict the numerics, and none should be sought. The conjectured large- m statement, $\delta(m) - \frac{1}{2} \sim (2\pi \text{Var}_4(m))^{-1/2}$ with $\text{Var}_4(m)$ as in Results item 3 (asymptotically equal to $\sigma_m^2 = \text{Var}(V_m)$ above), is the dissolution law of the Results section; its proof is the subject of Theorem 2 of the companion draft and is *not* proved in this paper.

Remark on constants. Plain Hoeffding (range form: $\mathbb{P}(S \leq -t) \leq \exp(-2t^2/\sum_i (\text{range}_i)^2)$ with $\text{range}_\gamma = 4a_\gamma$) gives $\exp(-1/(8A)) \leq \exp(-1/(4C(2m+1)^2))$, valid for all m as well; the arcsine moment-generating function is exactly I_0 , and $I_0(x) \leq e^{x^2/4}$ beats Hoeffding's lemma

($e^{x^2/2}$ for a $[-x, x]$ -valued variable) by a factor 2 in the exponent; no smallness restriction on m is needed anywhere.

Numerical consistency (deterministic Gil–Pelaez inversion, 511 ordinates plus Gaussianized tail; computational, not part of the proof): at $m = 0$, $1 - \delta = 4.073 \times 10^{-3}$ against the bounds $\min(0.156, e^{-1/(2C)} = 0.0406)$; at $m = -0.1$, 3.12×10^{-4} vs 6.7×10^{-3} ; at $m = -0.2$, 6.28×10^{-7} vs 1.36×10^{-4} ; at $m = -0.3$ the exact-MGF Chernoff value certifies $1 - \delta \leq 1.2 \times 10^{-15}$ vs the stated bound 2.0×10^{-9} . The bounds are (as they must be) weaker than the truth at every tested point. For comparison, the Hoeffding-form bound crosses below Chebyshev at $(2m + 1)^2 \approx 0.744$, i.e. $m \approx -0.069$; the I_0 -form bound is stronger than Chebyshev for all m . $\square$

Proof of (iv). Immediate from (i)–(iii) together with the cited endpoint results: [1] and [9] at $m = 0$ (our deterministic recomputation agrees to six decimals), and [2, 3, 14] at $m = -\frac{1}{2}$ for the unlogged race as quoted in the statement. We emphasize the logical status: $\delta(m)$ is defined for $m \in (-\frac{1}{2}, \infty)$ and $m = -\frac{1}{2}$ is not a value of the family; the connection asserted is $\delta(m) \rightarrow 1$ as $m \rightarrow -\frac{1}{2}^+$ (proved in (iii) under GRH+LI) matching the almost-sure single-sign regime established at $m = -\frac{1}{2}$ by different methods and under different hypotheses (DRH, resp. GRH variants, no LI). For the θ -form race at the threshold itself, the prime-square mechanism of Lemma 3 produces a $\frac{1}{2} \log x$ drift ($\sum_{p \leq \sqrt{x}} p^{-1} \log p \sim \frac{1}{2} \log x$); the $\frac{1}{2} \log \log x$ drift of [2, 3] belongs to the unlogged race $D_{-1/2}$, and the two are compatible by partial summation. We do not claim the $m = -\frac{1}{2}$ results, nor any new statement at $m = -\frac{1}{2}$. At the right end of the family we prove existence, continuity, and $\frac{1}{2} < \delta(m) < 1$ for every m , but no statement about the rate of approach to $\frac{1}{2}$; the decrease of the computed values in Table 1 is numerical observation, not a theorem of this paper. $\square$

Proposition 1 (transfer to the unlogged races). *Assume $GRH(\chi_4)$ and fix $m \in (-\frac{1}{2}, \infty)$. Let $D_m(x) = \sum_{p \leq x, p \equiv 3(4)} p^m - \sum_{p \leq x, p \equiv 1(4)} p^m$ and*

$$E'_m(x) = (2m + 1) x^{-(m+1/2)} (\log x) D_m(x).$$

Then $u \mapsto E'_m(e^u)$ has the same limiting distribution ν_m as $E_m(e^u)$; consequently, under $LI(\chi_4)$, all of (ii)–(iii) hold verbatim for the unlogged race, with the same $\delta(m)$.

Proof. Partial summation: $D_m(x) = \int_2^x (\log t)^{-1} dR_m(t) = \frac{R_m(x)}{\log x} + \int_2^x \frac{R_m(t)}{t \log^2 t} dt$. Insert the expansion of R_m from Lemmas 2–3 (truncation T , valid uniformly in $t \leq x$),

$$R_m(t) = \frac{t^{m+1/2}}{2m+1} + \sum_{|\gamma| \leq T} \frac{t^{\rho+m}}{\rho+m} + F(t, T),$$

$$F(t, T) \ll_m \frac{t^{m+1} \log^2(tT)}{T} + t^{m+1/3} \log t + t^{m+1/2} e^{-c\sqrt{\log t}} + (1 + t^m) \log t + \log^2 t,$$

the fourth term being the full-range floor of (2) (for $m \leq 0$ it is $\leq 2 \log t$ and merges with the last term, giving back the earlier display).

Write $s = \rho + m = m + \frac{1}{2} + i\gamma$, so $|s| \asymp \gamma$ and $\operatorname{Re} s = m + \frac{1}{2} > 0$.

Bias term. With $a = m + \frac{1}{2}$, two integrations by parts give $\int_2^x t^{a-1} \log^{-2} t dt = \frac{x^a}{a \log^2 x} (1 + O_a(1/\log x)) + O_a(1)$, whence

$$\frac{1}{2m+1} \left(\frac{x^{m+1/2}}{\log x} + \int_2^x \frac{t^{m+1/2}}{t \log^2 t} dt \right) = \frac{x^{m+1/2}}{(2m+1) \log x} \left(1 + O_m \left(\frac{1}{\log x} \right) \right).$$

Zero terms. For each zero, one integration by parts gives

$$\frac{1}{s} \int_2^x t^{s-1} \log^{-2} t \, dt = \frac{x^s}{s^2 \log^2 x} - \frac{2^s}{s^2 \log^2 2} + \frac{2}{s^2} \int_2^x t^{s-1} \log^{-3} t \, dt,$$

so the ρ -contribution to $D_m(x)$ is $\frac{x^s}{s \log x} (1 + O(\frac{1}{|s| \log x})) + O(|s|^{-2}) + O_m(|s|^{-2} x^{m+1/2} \log^{-3} x)$. The *summability in γ is what makes the bookkeeping close*: summing over $|\gamma| \leq T$, the three error families contribute, after multiplication by the normalizer $(2m+1)x^{-(m+1/2)} \log x$,

$$\left| \sum_{0 < \gamma \leq T} r_\gamma(m) e^{i\gamma u} O\left(\frac{1}{\gamma u}\right) \right| \ll_m \frac{1}{u} \sum_\gamma \frac{1}{\gamma^2} \ll_m \frac{1}{u}, \quad u e^{-(m+1/2)u} \sum_\gamma \gamma^{-2} \ll_m u e^{-(m+1/2)u}, \quad \frac{1}{\log^2 x} \sum_\gamma \gamma^{-2} \ll$$

all uniform in T (here $\sum_\gamma a_\gamma(m)/\gamma \ll_m \sum \gamma^{-2} < \infty$ is the key convergence).

Truncation/PNT term. Each component of F passes through $\frac{R_m(x)}{\log x}$ and $\int_2^x F(t, T) t^{-1} \log^{-2} t \, dt$ with at most one extra factor $\log$. For the components carrying a power t^a with $a > 0$ ($a = m+1$ and $a = m + \frac{1}{2}$ always; $a = m + \frac{1}{3}$ when $m > -\frac{1}{3}$), splitting the integral at $\sqrt{x}$ gives $\int_2^x t^{a-1} \log^{-2} t \, dt \ll_a x^a \log^{-2} x$, the PNT component retaining its factor $e^{-c\sqrt{\log t}}$ (decreasing, so bounded on $[\sqrt{x}, x]$ by $e^{-(c/\sqrt{2})\sqrt{\log x}}$). For $m \leq -\frac{1}{3}$ the $k \geq 3$ component has $m + \frac{1}{3} \leq 0$, so $t^{m+1/3} \log t \leq (\log 2)^{-1} \log^2 t$ for $t \geq 2$ and it is absorbed by the $\log^2 t$ component (this case must be separated: for $a \leq 0$ the integral $\int_2^x t^{a-1} \log^{-2} t \, dt$ is $\asymp_m 1$, not $\ll x^a \log^{-2} x$). The $\log^2 t$ component gives $\int_2^x t^{-1} \log^{-2} t \, dt = \log x - \log 2$; and the new floor $(1+t^m) \log t$ contributes $\int_2^x (1+t^m) t^{-1} \log^{-1} t \, dt \ll_m \log \log x + 1 + x^m / \log x$. After multiplication by the normalizer $(2m+1)x^{-(m+1/2)} \log x$, all of these are covered by the display below (the floor's contribution is $\ll_m u^2 e^{-\kappa_m u}$).

Collecting, for $u \geq 2$, $T \geq 2$,

$$E'_m(e^u) = 1 + \operatorname{Re} \sum_{0 < \gamma \leq T} r_\gamma(m) e^{i\gamma u} \mathcal{E}'(u, T), \quad \mathcal{E}'(u, T) \ll_m \frac{u e^{u/2} \log^2(e^u T)}{T} + \frac{1}{u} + u^2 e^{-\kappa_m u} + u e^{-u/6} + u e^{-c\sqrt{u}},$$

with $\kappa_m = \min(m + \frac{1}{2}, \frac{1}{2})$ as in the proof of (i), and at $T = e^Y$, $u \in [2, Y]$ each term's Cesàro mean square tends to 0 exactly as in (i) (the first is $\ll Y^5 e^{-Y}$ after integration; $1/u$ contributes $\frac{1}{Y} \int_2^Y u^{-2} du \ll 1/Y$; the remaining three are fixed $L^2(0, \infty)$ functions of u since $\kappa_m > 0$). Thus $E'_m(e^u)$ satisfies the same [7] hypotheses with the *same* constant 1 and the *same* coefficients $r_\gamma(m)$, so its limiting distribution is again ν_m . At $m = 0$ this recovers the classical passage of [1] between θ - and π -races.

□

Remark 2. *Two further points of independent interest. (1) Without LI, part (i) still gives existence of ν_m , and [7, Thm. 1.14] (Parseval, literally their eq. (1.24) with $c = 1$) evaluates the mean square of the normalized race: $\lim_Y \frac{1}{Y} \int_0^Y E_m(e^u)^2 du = 1 + \frac{1}{2} \sum_\gamma |r_\gamma|^2 = 1 + \sum_\gamma 2a_\gamma(m)^2 \leq 1 + C(2m+1)^2$, so the spread about the unit bias collapses quadratically in $(2m+1)$ under GRH alone. We do not pursue LI-free density statements (for which see [14], and [8] for the possible-atom phenomenology), because without LI ν_m could in principle carry an atom at 0 and the density interpretation requires care. (2) The bounds of (iii) hold with $\sigma_m^2 = \sum_\gamma 2a_\gamma(m)^2$ in place of $C(2m+1)^2$ throughout, which is numerically slightly sharper (at $m = 0$: 0.15557 vs 0.15603); we state the $C(2m+1)^2$ form because the m -dependence is transparent.*

Monotonicity in the weight

Every computed value in this note is consistent with a strict monotonicity that the dissolution asymptotics alone do not give:

Conjecture 1. *For $q \in \{3, 4\}$ the map $m \mapsto \delta_q(m)$ is strictly decreasing on $(-\frac{1}{2}, \infty)$.*

Three remarks on its status. (a) *Numerical evidence.* Deterministic Gil–Pelaez inversion with certified enclosures — the unseen-zero tail variance evaluated in closed form from the completed L -function (the Hadamard-product identity behind the evaluation of C), the tail’s departure from Gaussianity bounded by a proved quartic estimate, truncation and quadrature budgeted into the interval radius; every enclosure width below 5×10^{-7} — on the 30-point grid

$m \in \{-0.2, -0.15, -0.1, -0.05, 0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12, 15, 20, 25, 30, 40, 50,$

... yields enclosures that are pairwise disjoint and decreasing at all 29 consecutive pairs, for both moduli; the smallest gap between adjacent enclosures is 2.9×10^{-5} for modulus 4 and 8.6×10^{-7} for modulus 3 (both at the leftmost pair). The enclosures are rigorous modulo the stated computational budget (IEEE-double quadrature with step-doubling control; 40-digit floating-point evaluation of the closed-form variance; no interval arithmetic is claimed). Left of the grid the inversion loses resolution: at $m = -0.2$ itself the enclosures give $1 - \delta_4 \in [4.8, 7.7] \times 10^{-7}$ and $1 - \delta_3 \leq 2.5 \times 10^{-7}$, and on $(-\frac{1}{2}, -0.2)$ the only control we have is the Chernoff bound of Theorem 1(iii) (stated for χ_4 ; the same argument applies verbatim mod 3), a monotone upper envelope for $1 - \delta$ which is consistent with — but does not imply — monotonicity of δ itself.

(b) *Why the conjecture is plausible.* With $M = m + \frac{1}{2}$, every amplitude is strictly increasing in the weight:

$$\frac{\partial a_\gamma}{\partial M} = \frac{2\gamma^2}{(M^2 + \gamma^2)^{3/2}} > 0,$$

so as m grows each noise component of $X_m = 1 + \sum_{\gamma>0} 2a_\gamma \cos \theta_\gamma$ strictly grows while the mean stays fixed at 1; the conjecture asserts that this pointwise growth of the noise always lowers $\mathbb{P}(X_m > 0)$.

(c) *Why it is not a theorem.* If the family were a dilation, $V_m = c(m)V$ with c increasing, monotonicity would be immediate; it is not — the ratio $a_\gamma(m')/a_\gamma(m)$ depends on γ (low zeros saturate near 2, high zeros still grow linearly in M). The natural tool is a peakedness ordering: for the symmetric variable $V_m = X_m - 1$ one has $\delta(m) = 1 - \frac{1}{2} \mathbb{P}(|V_m| \geq 1)$, so it would suffice that $V_{m'}$ is more peaked than V_m for $m' < m$ in the sense of Birnbaum [24]. Scaling up a single component always makes that component less peaked, but the classical results that push such comparisons through convolution — Birnbaum’s lemma [24] and Proschan’s theorem on convex combinations [25] — require symmetric *unimodal* summands (Proschan’s in the stronger log-concave form), and the arcsine components $2a_\gamma \cos \theta_\gamma$ are symmetric but not unimodal (their density is U-shaped and diverges at the endpoints). We know of no peakedness comparison valid for arcsine components, and the conjecture remains open: neither the Chernoff envelope near $m = -\frac{1}{2}$ nor the large- m dissolution asymptotics controls the difference $\delta(m) - \delta(m')$ at nearby weights.

3 The variance in closed form

The variance $\sigma_m^2 = \sum_{\gamma>0} 2a_\gamma(m)^2$ of the limiting variable X_m has so far entered this paper through a truncation: 511 computed ordinates plus a tail estimated from the zero-counting

density. This section removes the estimate. The variance is the value of an explicitly computable entire function of the weight, with three consequences: the constant C of Theorem 1 acquires a closed form (and its printed value a correction of 3.4×10^{-6}); the variance half of the dissolution law of the introduction becomes a theorem with an explicit constant term; and the Gaussianized tails of the numerical work become exactly computable quantities.

Throughout, $\Lambda(s) := (4/\pi)^{(s+1)/2} \Gamma(\frac{s+1}{2}) L(s, \chi_4)$ denotes the completed L -function and $\Xi(z) := \Lambda(\frac{1}{2} + z)$.

Proposition 2 (variance identity). *Assume GRH(χ_4). For every real $m > -\frac{1}{2}$, with $M = m + \frac{1}{2}$,*

$$\sigma_m^2 = \sum_{\gamma > 0} 2a_\gamma(m)^2 = 8M^2 \sum_{\gamma > 0} \frac{1}{M^2 + \gamma^2} = 4M \frac{\Lambda'}{\Lambda}(m+1). \quad (4)$$

GRH enters only by placing the zeros on the critical line, so that the ordinates γ are real; no linear-independence hypothesis is involved.

Proof. The first equality in (4) is the definition together with $a_\gamma(m)^2 = (2M)^2/(M^2 + \gamma^2)$. The content is the last equality.

Step 1: Ξ is entire of order 1 and even. Since χ_4 is primitive and nonprincipal, Λ is entire of order 1 [17, Chs. 9, 12]. For a primitive character $\chi \bmod q$ with $\chi(-1) = (-1)^a$, the functional equation reads $\Lambda(1-s, \bar{\chi}) = (i^a \sqrt{q}/\tau(\chi)) \Lambda(s, \chi)$ with $\tau(\chi) = \sum_{n \bmod q} \chi(n) e^{2\pi i n/q}$ [17, Ch. 9]. Here $q = 4$, $a = 1$ (as $\chi_4(-1) = \chi_4(3) = -1$), $\bar{\chi}_4 = \chi_4$, and the Gauss sum is a two-term computation:

$$\tau(\chi_4) = e^{2\pi i/4} - e^{2\pi i \cdot 3/4} = i - (-i) = 2i, \quad \text{so} \quad \frac{i^a \sqrt{q}}{\tau(\chi_4)} = \frac{2i}{2i} = 1,$$

i.e. the root number is $+1$ and $\Lambda(s) = \Lambda(1-s)$. Hence $\Xi(z) = \Xi(-z)$. (The script verifies the evenness numerically to 4×10^{-51} relative at several complex points.)

Step 2: no central zero. $L(\frac{1}{2}, \chi_4) = \sum_{k \geq 0} (-1)^k (2k+1)^{-1/2} > 0$ by the alternating series test (terms strictly decreasing to 0); numerically $L(\frac{1}{2}, \chi_4) = 0.667691457 \dots$. So $\Xi(0) = \Lambda(\frac{1}{2}) > 0$ (each factor of $\Lambda(\frac{1}{2})$ is positive), and no zero of Ξ sits at the origin.

Step 3: Hadamard product. The zeros of Ξ are exactly $z_\rho = \rho - \frac{1}{2}$ over the nontrivial zeros ρ of $L(s, \chi_4)$; under GRH and Step 2 they are $\pm i\gamma$ with $\gamma > 0$ real, and the multiset of ordinates is symmetric under $\gamma \mapsto -\gamma$ because Λ is real on the real axis. By the Riemann–von Mangoldt formula $N(T, \chi_4) \ll T \log T$ [17, §16], $\sum_{\gamma > 0} \gamma^{-2} < \infty$. Hadamard's factorization for the order-1 function Ξ gives

$$\Xi(z) = e^{A+Bz} \prod_{\rho} \left(1 - \frac{z}{z_\rho}\right) e^{z/z_\rho},$$

the product converging absolutely and locally uniformly (indeed $\sum |z/z_\rho|^2 = |z|^2 \sum \gamma^{-2} < \infty$ locally uniformly, and $|\log\{(1-w)e^w\}| \leq |w|^2$ for $|w| \leq \frac{1}{2}$), so its factors may be grouped at will. Grouping $\pm i\gamma$ cancels the exponential factors pairwise:

$$\Xi(z) = e^{A+Bz} \prod_{\gamma > 0} \left(1 + \frac{z^2}{\gamma^2}\right).$$

The paired product is even in z ; evenness of Ξ then forces $e^{Bz} = e^{-Bz}$, i.e. $B = 0$ (alternatively, $\Xi'(0) = 0$ since Ξ is even and analytic, and the paired product has vanishing derivative at 0). Setting $z = 0$ identifies $e^A = \Xi(0) \neq 0$:

$$\frac{\Xi(z)}{\Xi(0)} = \prod_{\gamma>0} \left(1 + \frac{z^2}{\gamma^2}\right). \quad (5)$$

Step 4: logarithmic differentiation on the real axis. Under GRH all zeros of Ξ are purely imaginary and nonzero, so $\Xi > 0$ on $\mathbb{R}$ (it is real there, positive at 0, and nonvanishing). Taking logarithms in (5) at real $z = x$,

$$\log \frac{\Xi(x)}{\Xi(0)} = \sum_{\gamma>0} \log \left(1 + \frac{x^2}{\gamma^2}\right),$$

a series of nonnegative terms dominated on $|x| \leq A$ by A^2/γ^2 ; the termwise-differentiated series $\sum_{\gamma} 2x/(x^2 + \gamma^2)$ is dominated on $|x| \leq A$ by $2A/\gamma^2$. Both therefore converge uniformly on compact subsets of $\mathbb{R}$, and the standard theorem on differentiation of series (pointwise convergence plus locally uniform convergence of the derivative series) yields, for every real x ,

$$\frac{\Xi'}{\Xi}(x) = \sum_{\gamma>0} \frac{2x}{x^2 + \gamma^2}.$$

At $x = M > 0$, since $\Xi'/\Xi(M) = (\Lambda'/\Lambda)(M + \frac{1}{2}) = (\Lambda'/\Lambda)(m + 1)$,

$$4M \frac{\Lambda'}{\Lambda}(m + 1) = 8M^2 \sum_{\gamma>0} \frac{1}{M^2 + \gamma^2} = \sigma_m^2. \quad \square$$

Remark 3. *The right-hand side of (4) is computable to arbitrary precision with no zero data: $(\Lambda'/\Lambda)(s) = \frac{1}{2} \log \frac{4}{\pi} + \frac{1}{2} \psi\left(\frac{s+1}{2}\right) + (L'/L)(s, \chi_4)$, with $L(s, \chi_4) = 4^{-s}(\zeta(s, \frac{1}{4}) - \zeta(s, \frac{3}{4}))$. As a consistency check on both the identity and our ordinate list, the truncated zero sum $\sum_{\gamma \leq \gamma_{511}} 2a_{\gamma}^2$ plus the Riemann–von Mangoldt tail integral reproduces (4) at $m \in \{0, 1, 2, 3, 8, 20, 100\}$ within the expected fluctuation of the zero-counting remainder (3.2×10^{-6} at $m = 0$, growing to 0.13 absolute — 1.5×10^{-4} relative — at $m = 100$), and $\log(\Xi(z)/\Xi(0)) - \sum_{\gamma \leq \gamma_{511}} \log(1 + z^2/\gamma^2)$ matches the corresponding tail integral $\int_{\gamma_{511}}^{\infty} \log(1 + z^2/t^2) dN(t)$ to 1.6×10^{-6} at $z = 1$.*

Corollary 1 (the constant C in closed form). *Assume GRH(χ_4). With G Catalan's constant,*

$$C = \sum_{\gamma>0} \frac{2}{\gamma^2} = (\log \Lambda)''(\tfrac{1}{2}) = \tfrac{1}{4} \psi'(\tfrac{3}{4}) + (\log L)''(\tfrac{1}{2}, \chi_4), \quad \psi'(\tfrac{3}{4}) = \pi^2 - 8G,$$

and numerically

$$C = 0.156029649889644884607682\dots, \quad \tfrac{1}{4} \psi'(\tfrac{3}{4}) = 0.635469911917901\dots, \quad (\log L)''(\tfrac{1}{2}, \chi_4) = -0.479440262028256\dots$$

This supersedes the zero-sum estimate $C = 0.156033$ of earlier drafts, whose excess $+3.4 \times 10^{-6}$ is the neglected fluctuation term flagged there as $O(10^{-5})$ (the unrounded zero-sum-plus-tail estimate is 0.1560328, an excess of 3.2×10^{-6} ; rounding to the printed six decimals accounts for the rest). All bounds stated with C only improve.

Proof. Differentiating $\sum_{\gamma} 2x/(x^2 + \gamma^2)$ termwise once more is legitimate on $|x| \leq 1$: the differentiated terms $2(\gamma^2 - x^2)/(\gamma^2 + x^2)^2$ are bounded there by $2/\gamma^2$ (numerator at most γ^2 , denominator at least γ^4 ; recall $\gamma_1 > 6$), giving uniform convergence. Evaluating at $x = 0$ yields $(\log \Xi)''(0) = \sum_{\gamma > 0} 2/\gamma^2 = C$, and $(\log \Xi)''(0) = (\log \Lambda)''(\frac{1}{2})$. Since the linear factor of $\log \Lambda(s) = \frac{s+1}{2} \log \frac{4}{\pi} + \log \Gamma(\frac{s+1}{2}) + \log L(s, \chi_4)$ has vanishing second derivative, $(\log \Lambda)''(\frac{1}{2}) = \frac{1}{4} \psi'(\frac{3}{4}) + (\log L)''(\frac{1}{2}, \chi_4)$; the argument $\frac{3}{4}$ is $(s+1)/2$ at $s = \frac{1}{2}$, and $\log L$ is well defined near $s = \frac{1}{2}$ by Step 2 above. Finally $\psi'(\frac{1}{4}) + \psi'(\frac{3}{4}) = \pi^2/\sin^2(\pi/4) = 2\pi^2$ (reflection) and, writing $\psi'(x) = \sum_{n \geq 0} (n+x)^{-2}$, $\psi'(\frac{1}{4}) - \psi'(\frac{3}{4}) = 16 \sum_{n \geq 0} [(4n+1)^{-2} - (4n+3)^{-2}] = 16\beta(2) = 16G$, whence $\psi'(\frac{3}{4}) = \pi^2 - 8G$.

On the numerical claim, an honesty note: the closed form is exact, and its decimal evaluation is a routine high-precision computation (mpmath at 50 digits with guard digits; the routes $\frac{1}{4} \psi'(\frac{3}{4}) + (\log L)''(\frac{1}{2})$ and direct numerical $(\log \Lambda)''(\frac{1}{2})$ agree to 38 digits, stably under doubling of the working precision). We have not run certified interval arithmetic; the quantities are derivatives of entire functions evaluated far from any singularity. $\square$

Proposition 3 (rigorous variance asymptotic). *Assume $GRH(\chi_4)$. As $m \rightarrow \infty$,*

$$\sigma_m^2 = 2M \log \frac{4M}{2\pi} + 1 + \frac{1}{12M} + O\left(\frac{1}{M^2}\right). \quad (6)$$

More generally, let χ be a real primitive character mod q with $\chi(-1) = (-1)^a$, $a \in \{0, 1\}$, assume $GRH(\chi)$ and $L(\frac{1}{2}, \chi) \neq 0$, and set $\text{Var}_q(m) = \sum_{\gamma > 0} 2a_{\gamma}(m)^2$ over the ordinates of $L(s, \chi)$. Then

$$\text{Var}_q(m) = 2M \log \frac{qM}{2\pi} + (2a - 1) + \frac{1}{12M} + O\left(\frac{1}{M^2}\right), \quad (7)$$

with an absolute implied constant. In particular the constant term of the dissolution law of the introduction is +1 for odd real characters ($q = 3, 4$ included) and -1 for even ones.

Proof. Proposition 2 and its proof apply verbatim to χ : the completed function $\xi(s, \chi) = (q/\pi)^{(s+a)/2} \Gamma(\frac{s+a}{2}) L(s, \chi)$ is entire of order 1; a real primitive character has root number +1 — Gauss's evaluation of the sign of the quadratic Gauss sum, $\tau(\chi) = \sqrt{q}$ for $a = 0$ and $i\sqrt{q}$ for $a = 1$ (see [17, Ch. 2] for prime modulus; the script evaluates τ directly for the moduli 3, 4, 5 used in this paper) — so $\xi(s, \chi) = \xi(1-s, \chi)$; the hypothesis $L(\frac{1}{2}, \chi) \neq 0$ replaces Step 2 (for $q = 3, 4$ it holds unconditionally by grouping the Dirichlet series at $s = \frac{1}{2}$ into positive blocks); and

$$\text{Var}_q(m) = 4M \frac{\xi'}{\xi}(m+1, \chi) = 4M \left[\frac{1}{2} \log \frac{q}{\pi} + \frac{1}{2} \psi\left(\frac{m+1+a}{2}\right) + \frac{L'}{L}(m+1, \chi) \right]. \quad (8)$$

Digamma term. Stirling's series $\psi(x) = \log x - \frac{1}{2x} - \frac{1}{12x^2} + O(x^{-4})$, $x \rightarrow \infty$, at $x = \frac{m+1+a}{2} = \frac{M}{2} + \frac{2a+1}{4}$. For $a = 1$, $x = \frac{M}{2} (1 + \frac{3}{2M})$ and, expanding each ingredient to two orders,

$$\log x = \log \frac{M}{2} + \frac{3}{2M} - \frac{9}{8M^2} + O(M^{-3}), \quad \frac{1}{2x} = \frac{1}{M} - \frac{3}{2M^2} + O(M^{-3}), \quad \frac{1}{12x^2} = \frac{1}{3M^2} + O(M^{-3}),$$

so that, with $-\frac{9}{8} + \frac{3}{2} - \frac{1}{3} = \frac{1}{24}$,

$$\psi\left(\frac{m+2}{2}\right) = \log \frac{M}{2} + \frac{1}{2M} + \frac{1}{24M^2} + O(M^{-3}).$$

For $a = 0$, $x = \frac{M}{2}(1 + \frac{1}{2M})$ and the same computation ($-\frac{1}{8} + \frac{1}{2} - \frac{1}{3} = \frac{1}{24}$ again) gives $\psi(\frac{m+1}{2}) = \log \frac{M}{2} - \frac{1}{2M} + \frac{1}{24M^2} + O(M^{-3})$. Multiplying by $4M \cdot \frac{1}{2}$ and adding $2M \log \frac{q}{\pi}$,

$$4M \left[\frac{1}{2} \log \frac{q}{\pi} + \frac{1}{2} \psi \left(\frac{m+1+a}{2} \right) \right] = 2M \log \frac{qM}{2\pi} + (2a-1) + \frac{1}{12M} + O(M^{-2}),$$

using $\frac{q}{\pi} \cdot \frac{M}{2} = \frac{qM}{2\pi}$; the sign of the $\pm \frac{1}{2M}$ term is the sole source of the parity-dependent constant $2a-1$.

The L'/L term is exponentially small. For $m \geq 1$,

$$\left| \frac{L'}{L}(m+1, \chi) \right| = \left| \sum_{n \geq 2} \Lambda(n) \chi(n) n^{-(m+1)} \right| \leq 2^{-(m-1)} \sum_{n \geq 2} \Lambda(n) n^{-2} \leq 0.57 \cdot 2^{-(m-1)}$$

(using $n^{-(m+1)} \leq n^{-2} \cdot 2^{-(m-1)}$ for $n \geq 2$ and $\sum_n \Lambda(n) n^{-2} = -\zeta'(2) = 0.56996\dots$), so $4M|L'/L(m+1, \chi)| \ll M2^{-m}$, absorbed into $O(M^{-2})$ (with an absolute constant, uniformly in q). For χ_4 the decay is in fact $3^{-(m+1)} \log 3 (1+o(1))$, since $\chi_4(2^k) = 0$ and the leading surviving term is $n = 3$ with $\chi_4(3) = -1$; the script confirms $|L'/L(21, \chi_4)| = 1.050 \times 10^{-10} = 3^{-21} \log 3$ to three digits. Combining the three displays proves (7), and $q = 4$, $a = 1$ gives (6). $\square$

Remark 4. (1) Numerically, $M(\sigma_m^2 - 2M \log \frac{4M}{2\pi} - 1)$ equals 0.080845, 0.082709, 0.083083 at $m = 100, 400, 1000$, converging to $\frac{1}{12} = 0.08333\dots$ exactly as (6) requires; for the even real character mod 5 (where $L(\frac{1}{2}, \chi_5) = 0.23175\dots \neq 0$ numerically), the same computation gives $\text{Var}_5(m) - 2M \log \frac{5M}{2\pi} \rightarrow -1$, confirming $c_0(q, 0) = -1$. (2) The law is asymptotic, not a bound: at $m = 1$ the main term $2M \log \frac{4M}{2\pi} = -0.138$ is negative and useless (Table 2). (3) What is not proved in this section: the density half of the dissolution law, $\delta_q(m) - \frac{1}{2} \sim (2\pi \text{Var}_q(m))^{-1/2}$. Proposition 3 certifies only its variance input.

Remark 5 (certified tails for the numerical densities). The Gil-Pelaez and Monte Carlo computations of Sections 1 and 5 replace the unseen zeros $\gamma > \Gamma := \gamma_{511} = 639.9157\dots$ by a Gaussian whose variance was previously estimated from the zero-counting density. By Proposition 2 that variance is now exact:

$$\sigma_{\text{tail}}^2(\Gamma) = \sum_{\gamma > \Gamma} 2a_\gamma(m)^2 = 4M \frac{\Lambda'}{\Lambda}(m+1) - \sum_{\gamma \leq \Gamma} 2a_\gamma(m)^2,$$

the subtracted sum being exact up to the 25-digit accuracy of the computed ordinates (the resulting uncertainty is below 10^{-20}). Table 2 gives the exact variances against the law of the introduction. The smoothed tail estimate used previously deviates from the exact tail by at most 1.5×10^{-4} relative to σ_m^2 over the computed range (worst at $m = 100$, where the tail carries 16.7% of the variance). Re-running the Gil-Pelaez inversion with the exact tail variances shifts the entries of Table 1 by at most 3.3×10^{-6} (worst at $q = 3$, $m = 1$): up to three units in their sixth printed decimal — well within the 10^{-4} Monte Carlo cross-check, though not strictly below the printed resolution, so the last digit of a few entries moves — and the tail-variance estimation error is now exactly zero. For the record, the identity also gives $\sigma_0^2 = 0.1555679799235859\dots$ (the rounded 0.15557 quoted in Section 2 is unaffected).

m	σ_m^2 (exact, (4))	$2M \log \frac{4M}{2\pi}$	$\sigma_m^2 - 2M \log \frac{4M}{2\pi} - 1$	$\sigma_{\text{tail}}^2(\gamma_{511})$	tail share
1	1.368555	-0.138353	0.506908	0.031353	2.3%
3	6.776846	5.608262	0.168585	0.170697	2.5%
8	29.712468	28.704219	0.008249	1.006718	3.4%
20	106.326000	105.322529	0.003470	5.854189	5.5%
100	836.874384	835.873579	0.000804	139.712453	16.7%

Table 2: Exact variances from the identity (4) against the paper’s dissolution-law main term, modulus 4. The residual column converges to 0 like $\frac{1}{12M}$ once the constant $c_0 = 1$ of Proposition 3 is subtracted; at small m it is dominated by the (exponentially small in m , but not yet small) L'/L term and higher Stirling corrections. The last columns show the exact variance of the Gaussianized tail beyond the 511th ordinate and its share of σ_m^2 .

4 The dissolution law: from heuristic to theorem

The introduction stated the dissolution law $\delta_q(m) - \frac{1}{2} \sim (2\pi \text{Var}_q(m))^{-1/2}$ as a Gaussian-comparison heuristic with numerical confirmation, and Proposition 3 made its variance input $\text{Var}_q(m) = 2M \log \frac{qM}{2\pi} + O(1)$ rigorous. This section proves the density half, with a rate, and in fact with a proved second-order term. The method is direct Fourier inversion: the limiting variable is $1 + V$ with V a sum of independent scaled cosines, its characteristic function is a product of Bessel factors $J_0(2a_\gamma t)$, and

$$\delta_q(m) - \frac{1}{2} = \frac{1}{\pi} \int_0^\infty \Phi_m(t) \frac{\sin t}{t} dt, \quad \Phi_m(t) = \prod_{\gamma>0} J_0(2a_\gamma(m)t).$$

Near $t = 0$ the product is Gaussian to high accuracy, with variance σ_m^2 and a fourth-cumulant correction governed by $S_4(m) := \sum_{\gamma>0} (2a_\gamma(m))^4$; away from $t = 0$ the $\gtrsim M \log M$ zeros in the window $(M, 2M]$, each carrying an amplitude $a_\gamma \in [2/\sqrt{5}, \sqrt{2}]$ bounded away from 0 and ∞ , force superexponential decay. Both σ_m^2 and $S_4(m)$ turn out to be exact values of the completed L -function (Proposition 2 and Lemma 6), so every coefficient below is a closed form.

Setting and statement

Let χ be a real primitive character mod $q \geq 3$, $\chi(-1) = (-1)^a$ with $a \in \{0, 1\}$, and let $\xi(s, \chi) = (q/\pi)^{(s+a)/2} \Gamma(\frac{s+a}{2}) L(s, \chi)$ be its completed L -function. We assume $\text{GRH}(\chi)$ and $L(\frac{1}{2}, \chi) \neq 0$ (for $q = 3, 4$ the nonvanishing is unconditional, as noted in the proof of Proposition 3); γ runs over the positive ordinates of $L(s, \chi)$, listed with multiplicity. For $m > 0$ put $M = m + \frac{1}{2}$,

$$a_\gamma(m) = \frac{2M}{\sqrt{M^2 + \gamma^2}}, \quad X_{m,\chi} = 1 + V_m, \quad V_m = \sum_{\gamma>0} 2a_\gamma(m) \cos \theta_\gamma,$$

with θ_γ i.i.d. uniform on $[0, 2\pi)$ (the series converges a.s. and in L^2 , as in Section 2), and

$$\delta_q(m) := \mathbb{P}(X_{m,\chi} > 0), \quad \sigma_m^2 := \sum_{\gamma>0} 2a_\gamma(m)^2, \quad S_4(m) := \sum_{\gamma>0} (2a_\gamma(m))^4.$$

For $q = 4$, Theorem 1(ii) identifies $\delta_4(m)$, under $\text{LI}(\chi_4)$, with the logarithmic density $\delta(m)$ of the set where the weighted race R_m is positive; for other moduli $\delta_q(m)$ is, in this paper, a

statement about the limiting random variable only (see the Scope remark at the end of the section). Note that the theorem below does *not* assume LI: independence of the phases is built into the definition of $X_{m,\chi}$, and LI is what transfers the conclusion to the race.

Theorem 2 (dissolution law). *Let χ be a real primitive character mod q ; assume $GRH(\chi)$ and $L(\frac{1}{2}, \chi) \neq 0$. Then, with $\sigma_m^2 = 4M \frac{\xi'}{\xi}(m+1, \chi)$ as in (8) and $S_4(m)$ as in Lemma 6, as $m \rightarrow \infty$:*

$$(i) \quad \delta_q(m) - \frac{1}{2} = \frac{1}{\sqrt{2\pi\sigma_m^2}} \left(1 + O\left(\frac{1}{M \log M}\right) \right);$$

$$(ii) \quad \text{more precisely, } \delta_q(m) - \frac{1}{2} = \frac{1}{\sqrt{2\pi\sigma_m^2}} \left(1 - \frac{1}{6\sigma_m^2} - \frac{3S_4(m)}{64\sigma_m^4} + O\left(\frac{1}{(M \log M)^2}\right) \right);$$

(iii) consequently, by Proposition 3,

$$\delta_q(m) - \frac{1}{2} = \left(4\pi M \log \frac{qM}{2\pi} \right)^{-1/2} \left(1 + O\left(\frac{1}{M \log M}\right) \right).$$

All implied constants depend on q alone; they are effectively computable in principle and we have not optimized them. Nothing is claimed uniformly in q .

Corollary 2 (the race itself, $q = 4$). *Assume $GRH(\chi_4)$ and $LI(\chi_4)$. Then the logarithmic density $\delta(m)$ of $\{x \geq 2 : R_m(x) > 0\}$ of Theorem 1 satisfies (i)–(iii) of Theorem 2 verbatim. In particular $\delta(m) \rightarrow \frac{1}{2}$ as $m \rightarrow \infty$ at the rate $(4\pi M \log \frac{2M}{\pi})^{-1/2}$, closing the gap left open in Theorem 1(iii).*

Corollary 3 (closed second order). *Under the hypotheses of Theorem 2, with $L := \log \frac{qM}{2\pi}$,*

$$\delta_q(m) - \frac{1}{2} = \frac{1}{\sqrt{2\pi\sigma_m^2}} \left(1 - \frac{11}{24ML} + O\left(\frac{1}{ML^2}\right) \right) \quad (m \rightarrow \infty).$$

The relative error in (i) is thus not just $O(1/(M \log M))$ but asymptotically equal to $-\frac{11}{24}(ML)^{-1}$: the density approaches the Gaussian prediction from *below*, as every entry of Table 3 shows.

We assemble the proof from four lemmas: elementary Bessel estimates (Lemma 4); a count of zeros in the dyadic window $(M, 2M]$ (Lemma 5); the closed form and asymptotics of S_4 (Lemma 6); and the inversion formula (Lemma 7) together with the tail bound that justifies it (Lemma 8).

Lemmas

Lemma 4 (Bessel facts). *Let $J_0(z) = \sum_{k \geq 0} (-1)^k (z^2/4)^k / (k!)^2$. For real z :*

(J1) $|J_0(z)| \leq 1$, with strict inequality for $z \neq 0$; moreover for every $z_0 > 0$,

$$\rho(z_0) := \sup_{z \geq z_0} |J_0(z)| < 1.$$

(J2) $|J_0(z)| \leq \frac{8}{\pi} z^{-1/2}$ for all $z > 0$.

(J3) For $0 \leq z \leq 1$: $J_0(z) \geq \frac{3}{4}$ and $|\log J_0(z) + \frac{z^2}{4}| \leq \frac{z^4}{17}$.

(J4) For $0 \leq z \leq 1$: $|\log J_0(z) + \frac{z^2}{4} + \frac{z^4}{64}| \leq \frac{z^6}{80}$.

Proof. Throughout we use the integral representation

$$J_0(z) = \frac{2}{\pi} \int_0^{\pi/2} \cos(z \sin \varphi) d\varphi = \frac{2}{\pi} \int_0^1 \frac{\cos(zu)}{\sqrt{1-u^2}} du, \quad (9)$$

obtained by expanding $\cos(z \sin \varphi)$ into its (uniformly convergent) power series and integrating termwise with Wallis's formula $\int_0^{\pi/2} \sin^{2k} \varphi d\varphi = \frac{\pi}{2} \binom{2k}{k} 4^{-k}$, which reproduces the defining series; the second form is the substitution $u = \sin \varphi$.

(J1) The first form of (9) gives $|J_0(z)| \leq 1$. If $z \neq 0$, then $\varphi \mapsto z \sin \varphi$ is injective on $[0, \pi/2]$, so $\cos(z \sin \varphi) = \pm 1$ for at most finitely many φ , and by continuity $|J_0(z)| < 1$ strictly. For the supremum: by (J2), $|J_0(z)| \leq \frac{1}{2}$ for $z \geq 26 > (16/\pi)^2$; on the compact interval $[z_0, 26]$ the continuous function $|J_0|$ attains a maximum, which is < 1 by the strict inequality just proved. Hence $\rho(z_0) = \max(\max_{[z_0, 26]} |J_0|, \frac{1}{2}) < 1$ (for $z_0 \geq 26$ the compact piece is vacuous and $\rho(z_0) \leq \frac{1}{2}$ directly). (No numerical value of ρ is used in the proofs; for orientation, $\rho(1/(2\sqrt{5})) = J_0(1/(2\sqrt{5})) = 0.98754$, computed in the verification script.)

(J2) For $0 < z < 1$ the bound exceeds $8/\pi > 1$ and is trivial. For $z \geq 1$ set $\delta = 1/z \in (0, 1]$ and split (9) at $u = 1 - \delta$. The singular piece:

$$\int_{1-\delta}^1 \frac{du}{\sqrt{1-u^2}} \leq \int_{1-\delta}^1 \frac{du}{\sqrt{1-u}} = 2\sqrt{\delta}.$$

The oscillatory piece, integrating by parts ($\cos(zu) = \frac{1}{z} \frac{d}{du} \sin(zu)$) and using $1-u^2 \geq \delta(2-\delta) \geq \delta$ at $u = 1 - \delta$ together with $\int_0^{1-\delta} u(1-u^2)^{-3/2} du = (1-u^2)^{-1/2} \Big|_0^{1-\delta} \leq \delta^{-1/2}$:

$$\left| \int_0^{1-\delta} \frac{\cos(zu)}{\sqrt{1-u^2}} du \right| \leq \frac{1}{z} \cdot \frac{1}{\sqrt{\delta}} + \frac{1}{z} \int_0^{1-\delta} \frac{u du}{(1-u^2)^{3/2}} \leq \frac{2}{z\sqrt{\delta}}.$$

With $\delta = 1/z$ both pieces are $2z^{-1/2}$, so $|J_0(z)| \leq \frac{2}{\pi} \cdot 4z^{-1/2} = \frac{8}{\pi} z^{-1/2}$.

(J3) For $|z| \leq 1$ the series terms $c_k = (z^2/4)^k / (k!)^2$ decrease strictly ($c_{k+1}/c_k = z^2/(4(k+1)^2) \leq \frac{1}{4}$), so the alternating series gives

$$1 - \frac{z^2}{4} \leq J_0(z) \leq 1 - \frac{z^2}{4} + \frac{z^4}{64}, \quad \text{hence } J_0(z) \geq \frac{3}{4},$$

and, writing $J_0 = 1 - w$ with $w = \frac{z^2}{4} - R$,

$$R := \sum_{k \geq 2} (-1)^k c_k \in [0, \frac{z^4}{64}], \quad w \in [\frac{z^2}{4} - \frac{z^4}{64}, \frac{z^2}{4}] \subset [0, \frac{1}{4}].$$

Then $\log J_0 + \frac{z^2}{4} = R - \sum_{j \geq 2} \frac{w^j}{j}$, and $\sum_{j \geq 2} \frac{w^j}{j} \leq \frac{w^2}{2(1-w)} \leq \frac{2}{3} w^2 \leq \frac{2}{3} \left(\frac{z^2}{4}\right)^2 = \frac{z^4}{24}$, so the total is at most $\frac{z^4}{64} + \frac{z^4}{24} = \frac{11z^4}{192} \leq \frac{z^4}{17}$.

(J4) Refine each piece by one order. First, $R - \frac{z^4}{64} = \sum_{k \geq 3} (-1)^k c_k$ has absolute value at most $c_3 = \frac{z^6}{2304}$. Next, $w^2 = \frac{z^4}{16} - \frac{z^2 R}{2} + R^2$, so

$$\log J_0 + \frac{z^2}{4} + \frac{z^4}{64} = \left(R - \frac{z^4}{64}\right) + \frac{z^2 R}{4} - \frac{R^2}{2} - \sum_{j \geq 3} \frac{w^j}{j},$$

and, for $|z| \leq 1$: $\frac{z^2 R}{4} \leq \frac{z^6}{256}$, $\frac{R^2}{2} \leq \frac{z^8}{8192} \leq \frac{z^6}{8192}$, and $\sum_{j \geq 3} \frac{w^j}{j} \leq \frac{w^3}{3(1-w)} \leq \frac{4}{9} \left(\frac{z^2}{4}\right)^3 = \frac{z^6}{144}$. The sum of the four bounds is $z^6 \left(\frac{1}{2304} + \frac{1}{256} + \frac{1}{8192} + \frac{1}{144}\right) < 0.0115 z^6 < \frac{z^6}{80}$. (The true limit ratio as $z \rightarrow 0$ is $\frac{1}{576}$; the script checks the claimed bounds on a grid in multiprecision, float64 being insufficient near $z = 0$.) $\square$

Lemma 5 (zeros in the dyadic window). *Unconditionally, with $N_\chi(T) := \#\{\gamma \in (0, T]\}$, there is $M_0(q)$ such that for all $M \geq M_0(q)$,*

$$N_0 := N_\chi(2M) - N_\chi(M) \geq \frac{M}{4\pi} \log M \quad (\geq 5).$$

Proof. For a primitive $\chi \bmod q$ the Riemann–von Mangoldt formula gives $N_\chi(T) = \frac{T}{2\pi} \log \frac{qT}{2\pi e} + O(\log qT)$ for $T \geq 2$ [17, §16] (Davenport counts zeros with $|\gamma| \leq T$; for real χ the multiset of ordinates is symmetric under $\gamma \mapsto -\gamma$, as used in Proposition 2, so the one-sided count is half). Hence

$$N_\chi(2M) - N_\chi(M) = \frac{M}{2\pi} \left(\log \frac{qM}{2\pi e} + 2 \log 2 \right) + O_q(\log M) \geq \frac{M}{2\pi} \log \frac{qM}{2\pi e} - C_q \log M,$$

which exceeds $\frac{M}{4\pi} \log M$ once $M \geq M_0(q)$. $\square$

Lemma 6 (fourth moment of the amplitudes). *Assume $GRH(\chi)$ and $L(\frac{1}{2}, \chi) \neq 0$. Then for every $m > -\frac{1}{2}$,*

$$S_4(m) = \sum_{\gamma > 0} \frac{256 M^4}{(M^2 + \gamma^2)^2} = 16 \sigma_m^2 - 64 M^2 (\log \xi)''(m+1, \chi), \quad (10)$$

and, as $m \rightarrow \infty$,

$$S_4(m) = 32M \left(\log \frac{qM}{2\pi} - 1 \right) + 32(2a-1) + O_q\left(\frac{1}{M}\right). \quad (11)$$

Moreover, unconditionally on the size of m , $S_4(m) \leq 32 \sigma_m^2$ and $S_6(m) := \sum_\gamma (2a_\gamma)^6 \leq 16 S_4(m)$.

Proof. The trivial bounds are immediate from $(2a_\gamma)^2 \leq 16$ and $\sum_\gamma (2a_\gamma)^2 = 2\sigma_m^2$.

For (10): exactly as in the proof of the closed form for C (Corollary 1, applied to χ in place of χ_4 as in Proposition 3), the series $(\log \Xi)'(x) = \sum_{\gamma > 0} 2x/(x^2 + \gamma^2)$, $\Xi(z) := \xi(\frac{1}{2} + z, \chi)$, may be differentiated termwise on a neighbourhood of $x = M$: the differentiated terms $2(\gamma^2 - x^2)/(x^2 + \gamma^2)^2$ are dominated by $2/(x^2 + \gamma^2) \leq 2/\gamma^2$, uniformly on compacts. Hence

$$(\log \Xi)''(M) = \sum_{\gamma > 0} \frac{2(\gamma^2 - M^2)}{(M^2 + \gamma^2)^2},$$

and the identity is termwise algebra:

$$16 \cdot \frac{8M^2}{M^2 + \gamma^2} - 64M^2 \cdot \frac{2(\gamma^2 - M^2)}{(M^2 + \gamma^2)^2} = \frac{128M^2(M^2 + \gamma^2) - 128M^2\gamma^2 + 128M^4}{(M^2 + \gamma^2)^2} = \frac{256M^4}{(M^2 + \gamma^2)^2},$$

summed over $\gamma > 0$, using $\sigma_m^2 = \sum_\gamma 8M^2/(M^2 + \gamma^2)$ (Proposition 2, (8)) and $(\log \Xi)''(M) = (\log \xi)''(m+1, \chi)$.

For (11): $(\log \xi)''(s) = \frac{1}{4}\psi'(\frac{s+a}{2}) + (\frac{L'}{L})'(s, \chi)$. At $s = m + 1$, $x := \frac{m+1+a}{2} = \frac{M}{2} + \frac{2a+1}{4}$, and Euler–Maclaurin for $\psi'(x) = \sum_{n \geq 0} (n+x)^{-2}$ (comparison with $\int_0^\infty (t+x)^{-2} dt = 1/x$; correction $f(0)/2 = 1/(2x^2)$; remainder at most $\frac{1}{12}|f'(0)| + \frac{1}{12} \int_0^\infty |f''| \leq \frac{1}{2x^3}$) gives $\psi'(x) = \frac{1}{x} + \frac{1}{2x^2} + O(x^{-3})$, whence

$$\frac{1}{4}\psi'(x) = \frac{1}{2M} \cdot \frac{1}{1 + \frac{2a+1}{2M}} + \frac{1}{2M^2} + O\left(\frac{1}{M^3}\right) = \frac{1}{2M} - \frac{2a+1}{4M^2} + \frac{1}{2M^2} + O\left(\frac{1}{M^3}\right) = \frac{1}{2M} + \frac{1-2a}{4M^2} + O\left(\frac{1}{M^3}\right),$$

so $64M^2 \cdot \frac{1}{4}\psi'(x) = 32M + 16(1-2a) + O(1/M)$. The Dirichlet series $(\frac{L'}{L})'(s, \chi) = \sum_{n \geq 2} \Lambda(n)\chi(n)(\log n)n^{-s}$ is bounded at $s = m + 1$ by $2^{-(m-1)} \sum_n \Lambda(n)(\log n)n^{-2} \ll 2^{-m}$, so $64M^2(\frac{L'}{L})'(m+1) \ll M^2 2^{-m} = O(1/M)$ eventually. Combining with $16\sigma_m^2 = 32M \log \frac{qM}{2\pi} + 16(2a-1) + O(1/M)$ (Proposition 3) yields (11):

$$S_4(m) = 32M \log \frac{qM}{2\pi} + 16(2a-1) - 32M - 16(1-2a) + O(1/M). \quad \square$$

Numerically (10) and (11) agree to the expected order: at $(q, m) = (4, 100)$, the closed form gives $S_4 = 10190.02$, the 511 ordinates plus Riemann–von Mangoldt tail give 10190.11, and (11) without its $O(1/M)$ gives 10189.98.

Lemma 7 (characteristic function and inversion). *Let $V_m = \sum_{\gamma > 0} 2a_\gamma(m) \cos \theta_\gamma$ as above. Then:*

- (a) *the law of V_m is symmetric, and its characteristic function is $\Phi_m(t) = \prod_{\gamma > 0} J_0(2a_\gamma(m)t)$, the limit of the partial products, which is real and even; moreover for every finite set F of positive ordinates, $|\Phi_m(t)| \leq \prod_{\gamma \in F} |J_0(2a_\gamma(m)t)|$;*
- (b) *if $\Phi_m \in L^1(\mathbb{R})$, then V_m has a bounded continuous even density and*

$$\delta_q(m) - \frac{1}{2} = \mathbb{P}(V_m > -1) - \frac{1}{2} = \frac{1}{\pi} \int_0^\infty \Phi_m(t) \frac{\sin t}{t} dt,$$

the integral converging absolutely.

Proof. (a) For a single summand, $\mathbb{E} e^{it \cdot 2a \cos \theta} = \frac{1}{2\pi} \int_0^{2\pi} e^{2iat \cos \theta} d\theta = J_0(2at)$: expand the exponential and integrate termwise (odd powers vanish; $\frac{1}{2\pi} \int_0^{2\pi} \cos^{2k} \theta d\theta = \binom{2k}{k} 4^{-k}$), recovering the series of J_0 ; in particular each summand is symmetric ($\theta \mapsto \theta + \pi$ preserves the uniform law and flips the sign), hence so are the partial sums $S_N = \sum_{n \leq N} 2a_{\gamma_n} \cos \theta_{\gamma_n}$ and their a.s. limit V_m . Independence gives the characteristic function of S_N as the N -th partial product; $S_N \rightarrow V_m$ a.s. implies convergence in distribution, so the characteristic functions converge pointwise to that of V_m , which is therefore the limit of the partial products, real (symmetry) and even. For the subset bound, take N large enough that the first N ordinates contain F ; all factors have modulus ≤ 1 by (J1), so the modulus of the N -th partial product is at most $\prod_{\gamma \in F} |J_0|$, and let $N \rightarrow \infty$.

(b) By the Fourier inversion theorem for integrable characteristic functions [26, Ch. XV.3], the law of V_m has the bounded continuous density $f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} \Phi_m(t) dt$, which is even since Φ_m is real and even. Hence, using symmetry and then Fubini (justified by $\int_{\mathbb{R}} \int_{-1}^1 |\Phi_m(t)| dx dt = 2\|\Phi_m\|_1 < \infty$),

$$\mathbb{P}(V_m > -1) = \frac{1}{2} + \frac{1}{2} \mathbb{P}(|V_m| < 1) = \frac{1}{2} + \frac{1}{2} \int_{-1}^1 f = \frac{1}{2} + \frac{1}{4\pi} \int_{\mathbb{R}} \Phi_m(t) \left(\int_{-1}^1 e^{-itx} dx \right) dt,$$

and $\int_{-1}^1 e^{-itx} dx = 2 \sin t/t$, so $\mathbb{P}(V_m > -1) - \frac{1}{2} = \frac{1}{2\pi} \int_{\mathbb{R}} \Phi_m(t) \frac{\sin t}{t} dt = \frac{1}{\pi} \int_0^\infty \Phi_m(t) \frac{\sin t}{t} dt$ by evenness. Finally $\delta_q(m) = \mathbb{P}(1 + V_m > 0) = \mathbb{P}(V_m > -1)$. $\square$

Lemma 8 (superexponential decay for $t \geq \frac{1}{8}$). *Assume $GRH(\chi)$. Let $\rho := \rho(1/(2\sqrt{5})) < 1$ and $\tilde{\rho} := \max(\rho, \frac{1}{2}) < 1$ (absolute constants, by (J1)). There is $M_0(q)$ such that for all m with $M \geq M_0(q)$,*

$$\int_{1/8}^\infty |\Phi_m(t)| \frac{|\sin t|}{t} dt \leq 17 \tilde{\rho}^{N_0} \leq \exp\left(-\frac{\log(1/\tilde{\rho})}{8\pi} M \log M\right),$$

with $N_0 = N_\chi(2M) - N_\chi(M)$ as in Lemma 5; in particular $\Phi_m \in L^1(\mathbb{R})$.

Proof. For $\gamma \in (M, 2M]$ we have $M^2 + \gamma^2 \in (2M^2, 5M^2]$, so

$$a_\gamma(m) = \frac{2M}{\sqrt{M^2 + \gamma^2}} \in \left[\frac{2}{\sqrt{5}}, \sqrt{2}\right), \quad 2a_\gamma(m)t \geq \frac{4}{\sqrt{5}}t.$$

By Lemma 7(a) with $F = \{\gamma \in (M, 2M]\}$ (a set of N_0 ordinates, nonempty for $M \geq M_0(q)$): (i) for $t \geq \frac{1}{8}$, every factor has argument $\geq \frac{4}{\sqrt{5}} \cdot \frac{1}{8} = \frac{1}{2\sqrt{5}}$, so by (J1) $|\Phi_m(t)| \leq \rho^{N_0}$; (ii) for any $t > 0$, by (J2),

$$|\Phi_m(t)| \leq \left(\frac{8}{\pi} \left(\frac{4t}{\sqrt{5}}\right)^{-1/2}\right)^{N_0} = \left(\frac{c_3}{\sqrt{t}}\right)^{N_0}, \quad c_3 := \frac{4 \cdot 5^{1/4}}{\pi} = 1.9040 \dots < 2,$$

so that $c_3/\sqrt{t} \leq \frac{1}{2}$ for $t \geq 16$. Using $|\sin t|/t \leq 1$ on $[\frac{1}{8}, 16]$ and $|\sin t| \leq 1$ beyond, and (for $M \geq M_0(q)$, by Lemma 5) $N_0 \geq 5$:

$$\int_{1/8}^{16} |\Phi_m| \frac{|\sin t|}{t} dt \leq 16 \rho^{N_0}, \quad \int_{16}^\infty |\Phi_m| \frac{dt}{t} \leq 2^{5-N_0} c_3^5 \int_{16}^\infty t^{-7/2} dt = 2^{5-N_0} c_3^5 \cdot \frac{2}{5 \cdot 1024} \leq \frac{2^{-N_0}}{2},$$

where the middle step keeps five factors $(c_3/\sqrt{t})^5$ and bounds the remaining $N_0 - 5$ by $2^{-(N_0-5)}$. The sum is at most $17 \tilde{\rho}^{N_0}$, and with $N_0 \geq \frac{M}{4\pi} \log M$ the final bound follows for $M \geq M_0(q)$ enlarged if necessary (to absorb the factor 17 into half the exponent). The same estimates with $|\sin t|/t$ replaced by 1 on $[\frac{1}{8}, 16]$ and by nothing beyond show $\int_{1/8}^\infty |\Phi_m| < \infty$; on $[0, \frac{1}{8}]$, $|\Phi_m| \leq 1$. Hence $\Phi_m \in L^1$. $\square$

Proof of Theorem 2

Proof. Throughout, $M \geq M_0(q)$ so that Lemmas 5 and 8 apply, and $G := (2\pi\sigma_m^2)^{-1/2}$. By Lemma 8, $\Phi_m \in L^1$, so Lemma 7(b) gives

$$\delta_q(m) - \frac{1}{2} = \frac{1}{\pi} \int_0^{1/8} \Phi_m(t) \frac{\sin t}{t} dt + O(e^{-c_q M \log M}), \quad c_q := \frac{\log(1/\tilde{\rho})}{8\pi}. \quad (12)$$

The Gaussian factorization on $[0, \frac{1}{8}]$. For $0 \leq t \leq \frac{1}{8}$ every argument satisfies $2a_\gamma(m)t \leq 4t \leq \frac{1}{2}$, since $a_\gamma \leq 2$. By (J3) each factor is $\geq \frac{3}{4} > 0$ and $\sum_\gamma |\log J_0(2a_\gamma t)| \leq \sum_\gamma (a_\gamma^2 t^2 + (2a_\gamma t)^4/17) < \infty$, so the limit of the partial products is

$$\Phi_m(t) = \exp\left(-\frac{\sigma_m^2 t^2}{2} + E(t)\right), \quad E(t) := \sum_{\gamma > 0} \left(\log J_0(2a_\gamma t) + \frac{(2a_\gamma t)^2}{4}\right), \quad (13)$$

using $\sum_{\gamma}(2a_{\gamma})^2 t^2/4 = \sigma_m^2 t^2/2$. By (J3), $|E(t)| \leq \frac{t^4}{17} S_4$, and since $S_4 \leq 32\sigma_m^2$ (Lemma 6) and $t \leq \frac{1}{8}$,

$$|E(t)| \leq \frac{t^2}{17} \cdot \frac{1}{64} \cdot 32\sigma_m^2 = \frac{\sigma_m^2 t^2}{34} \quad (0 \leq t \leq \frac{1}{8}), \quad (14)$$

so that $|\Phi_m(t)| \leq e^{-(8/17)\sigma_m^2 t^2}$ there: the Gaussian decay survives the correction uniformly.

Decomposition. Write, on $[0, \frac{1}{8}]$, $\Phi_m \frac{\sin t}{t} = e^{-\sigma^2 t^2/2} - e^{-\sigma^2 t^2/2} (1 - \frac{\sin t}{t}) + e^{-\sigma^2 t^2/2} (e^E - 1) \frac{\sin t}{t}$ (abbreviating $\sigma = \sigma_m$), so that

$$\frac{1}{\pi} \int_0^{1/8} \Phi_m \frac{\sin t}{t} dt = G - T_1 - T_2 + T_3, \quad (15)$$

with, using $\frac{1}{\pi} \int_0^{\infty} e^{-\sigma^2 t^2/2} dt = G$ and the moments $\frac{1}{\pi} \int_0^{\infty} t^{2k} e^{-\sigma^2 t^2/2} dt = (2k-1)!! G \sigma^{-2k}$:

T_1 : the completed Gaussian tail, $0 \leq T_1 = \frac{1}{\pi} \int_{1/8}^{\infty} e^{-\sigma^2 t^2/2} dt \leq \frac{8}{\pi \sigma^2} e^{-\sigma^2/128}$;

T_2 : the sin correction: since $0 \leq 1 - \frac{\sin t}{t} \leq \frac{t^2}{6}$ for all $t > 0$ (from $t - \frac{t^3}{6} \leq \sin t \leq t$),

$$0 \leq T_2 = \frac{1}{\pi} \int_0^{1/8} e^{-\sigma^2 t^2/2} \left(1 - \frac{\sin t}{t}\right) dt \leq \frac{1}{6\pi} \int_0^{\infty} t^2 e^{-\sigma^2 t^2/2} dt = \frac{G}{6\sigma^2};$$

T_3 : the Bessel correction: by (14) and $|e^w - 1| \leq |w|e^{|w|}$,

$$|T_3| \leq \frac{1}{\pi} \int_0^{1/8} e^{-\sigma^2 t^2/2} \cdot \frac{t^4 S_4}{17} e^{\sigma^2 t^2/34} dt \leq \frac{S_4}{17\pi} \int_0^{\infty} t^4 e^{-\frac{8}{17}\sigma^2 t^2} dt = \frac{3S_4}{8 \cdot 17\sqrt{\pi}} \left(\frac{17}{8}\right)^{5/2} \sigma^{-5} \leq \frac{S_4}{12\sigma^5} \leq \frac{G S_4}{4\sigma^4},$$

the last step by $\sigma^{-5} = \sqrt{2\pi} G \sigma^{-4}$ and $\sqrt{2\pi}/12 < \frac{1}{4}$.

Part (i). By Proposition 3 there is $M_1(q) \geq M_0(q)$ with $ML \leq \sigma_m^2 \leq 3ML$ for $M \geq M_1(q)$, where $L = \log \frac{qM}{2\pi} \asymp_q \log M$; and $S_4 \leq 33ML$ by (11). Hence

$$\frac{T_2}{G} \leq \frac{1}{6ML}, \quad \frac{|T_3|}{G} \leq \frac{33ML}{4M^2 L^2} \leq \frac{9}{ML}, \quad \frac{T_1 + O(e^{-c_q M \log M})}{G} \ll \sigma e^{-\sigma^2/128} + \sigma e^{-c_q M \log M} \leq \frac{1}{ML}$$

for M large, and (12)–(15) give $\delta_q(m) - \frac{1}{2} = G(1 + O_q(1/(ML)))$, which is (i).

Part (ii). Refine T_2 and T_3 by one order each. For T_2 : on $[0, \frac{1}{8}]$ the alternating series of $\frac{\sin t}{t}$ has decreasing terms, so $|1 - \frac{\sin t}{t} - \frac{t^2}{6}| \leq \frac{t^4}{120}$, and extending the integral to ∞ costs $O(e^{-\sigma^2/129})$:

$$T_2 = \frac{1}{6\pi} \int_0^{\infty} t^2 e^{-\sigma^2 t^2/2} dt + O\left(\frac{1}{120\pi} \int_0^{\infty} t^4 e^{-\sigma^2 t^2/2} dt\right) + O(e^{-\sigma^2/129}) = \frac{G}{6\sigma^2} + O(G\sigma^{-4}).$$

For T_3 : by (J4), $E(t) = -\frac{t^4}{64} S_4 + E_2(t)$ with $|E_2(t)| \leq \frac{t^6}{80} S_6 \leq \frac{t^6}{5} S_4$ (Lemma 6); and $e^E - 1 = E + \omega$ with $|\omega| \leq \frac{E^2}{2} e^{|E|} \leq \frac{t^8 S_4^2}{578} e^{\sigma^2 t^2/34}$ by (14). Hence

$$T_3 = -\frac{S_4}{64\pi} \int_0^{1/8} t^4 e^{-\sigma^2 t^2/2} \frac{\sin t}{t} dt + O\left(\frac{S_4}{5\pi} \int_0^{\infty} t^6 e^{-\sigma^2 t^2/2} dt\right) + O\left(\frac{S_4^2}{578\pi} \int_0^{\infty} t^8 e^{-\frac{8}{17}\sigma^2 t^2} dt\right),$$

and in the first integral $\frac{\sin t}{t} = 1 + O(t^2)$, so, with the moment values and extension to ∞ as before,

$$T_3 = -\frac{3S_4}{64} G \sigma^{-4} + O(G S_4 \sigma^{-6}) + O(G S_4^2 \sigma^{-8}).$$

Since $\sigma^2 \asymp_q ML$ and $S_4 \asymp_q ML$, the error terms $G \cdot O(\sigma^{-4} + S_4 \sigma^{-6} + S_4^2 \sigma^{-8})$ are all $G \cdot O_q((ML)^{-2})$, and the exponentially small terms T_1 , (12) are absorbed. Collecting,

$$\delta_q(m) - \frac{1}{2} = G \left(1 - \frac{1}{6\sigma_m^2} - \frac{3S_4}{64\sigma_m^4} + O_q((M \log M)^{-2}) \right).$$

Part (iii). By Proposition 3, $\sigma_m^2 = 2ML + O(1)$, so $(2\pi\sigma_m^2)^{-1/2} = (4\pi ML)^{-1/2}(1 + O(1/(ML)))$; combine with (i). $\square$

Proof of Corollary 2. $L(\frac{1}{2}, \chi_4) > 0$ unconditionally, so Theorem 2 applies to χ_4 ; and under $\text{GRH}(\chi_4) + \text{LI}(\chi_4)$, Theorem 1(ii) gives $\delta(m) = \nu_m((0, \infty)) = \mathbb{P}(X_{m, \chi_4} > 0) = \delta_4(m)$. $\square$

Proof of Corollary 3. Insert $\sigma_m^2 = 2ML + O(1)$ and $S_4 = 32M(L - 1) + O(1)$ (Proposition 3, Lemma 6) into (ii):

$$\frac{1}{6\sigma_m^2} = \frac{1}{12ML} + O\left(\frac{1}{M^2L^2}\right), \quad \frac{3S_4}{64\sigma_m^4} = \frac{96M(L-1)}{256M^2L^2} + O\left(\frac{1}{M^2L^2}\right) = \frac{3}{8ML} + O\left(\frac{1}{ML^2}\right),$$

and $\frac{1}{12} + \frac{3}{8} = \frac{11}{24}$; the error $O((ML)^{-2})$ of (ii) is also $O(1/(ML^2))$. $\square$

Remark 6 (numerical consistency). *Table 3 compares the theorem with deterministic Gil-Pelaez reference densities (computational, not part of any proof): 511 resp. 537 ordinates, unseen zeros Gaussianized with the exact tail variance of Remark 5, trapezoidal inversion with grid-halving and window-doubling stable to 10^{-9} . Both σ_m^2 and S_4 are evaluated by their closed forms (8), (10) — no zero data enters the predictions. Gaussianizing the unseen zeros biases the reference values themselves: on $t \leq \frac{1}{8}$ the T_3 estimate, applied to the tail factors alone, rigorously bounds the relative bias by $0.21 S_4^{\text{tail}}/\sigma_m^4$; on $t > \frac{1}{8}$ that estimate does not apply, and we instead bound the remaining bias by the same integrand with the computed $|\Phi_m|$ inserted, which is below 9×10^{-7} at every table point. The total — 9.9×10^{-6} at $(q, m) = (4, 100)$, less elsewhere in the table — is the accuracy floor of the comparison; the $t > \frac{1}{8}$ piece is a numerical evaluation, not a theorem, on the same footing as the reference densities themselves. The certified enclosures of Table 4, computed by an independent route, contain the (unrounded) reference values used here at all four table points (e.g. $0.51377836 \in [0.5137782134, 0.5137783837]$ at $(4, 100)$; the table entry 0.5137784 is that reference rounded to seven decimals and lies just above the enclosure's upper end, consistent with the one-sided Gaussianization bias of the reference noted above).*

Corollary 3 at $(4, 100)$: $-\frac{11}{24}(ML)^{-1} = -1.10 \times 10^{-3}$ against the observed -8.80×10^{-4} ; the difference, 2.2×10^{-4} , is consistent with the corollary's $O(1/(ML^2)) = O(5.8 \times 10^{-4})$ — at these heights $L \approx 4.2$ is still small, which is why the table works with (ii) rather than the corollary. The reference values $\delta_4(100) = 0.5137784$ and $\delta_3(60) = 0.5197175$ refine the six-decimal values of the introduction: 0.513778 in Table 1, and 0.519716 in the text beside it (the $(3, 60)$ point is not a table entry; its value shifts by 1.5×10^{-6} upon replacing the estimated tail variance of the original computation by the exact one).

(q, m)	σ_m^2	S_4	$\delta_q(m)$ (ref.)	$\frac{\delta_q(m) - \frac{1}{2}}{(2\pi\sigma_m^2)^{-1/2}} - 1$	$-\frac{1}{6\sigma_m^2} - \frac{3S_4}{64\sigma_m^4}$	residual
(4, 8)	29.7125	219.83	0.5718701	-1.80×10^{-2}	-1.73×10^{-2}	-7.4×10^{-4}
(4, 20)	106.3260	1061.32	0.5384553	-6.05×10^{-3}	-5.97×10^{-3}	-7.9×10^{-5}
(4, 100)	836.8744	10190.02	0.5137784	-8.80×10^{-4}	-8.81×10^{-4}	$+7.0 \times 10^{-7}$
(3, 60)	407.9701	4607.56	0.5197175	-1.71×10^{-3}	-1.71×10^{-3}	-4.9×10^{-6}

Table 3: The dissolution law against deterministic reference densities. Column 5 is the observed relative deviation from the leading term of Theorem 2(i); column 6 is the second-order term of (ii); the residual — the relative error of the two-term prediction, equal to (col. 5 – col. 6)/(1 + col. 6) — shrinks like $(M \log M)^{-2}$, as (ii) requires — indeed at (4, 100) it is below the 9.9×10^{-6} modelling floor of the reference value, i.e. (ii) is confirmed there to the full accuracy of the computation. The $m = 100$ row is the consistency check promised in the introduction: $(2\pi \cdot 836.8744)^{-1/2} = 0.0137905$ against $\delta_4(100) - \frac{1}{2} = 0.0137784$, a relative gap of 8.8×10^{-4} , against the theorem’s error scale $1/(M \log \frac{qM}{2\pi}) = 1/418 = 2.4 \times 10^{-3}$.

Remark 7 (scope, and what is not proved). (1) Race interpretation. For $q = 4$, Corollary 2 makes the dissolution law a statement about the weighted prime race itself. For $q = 3$ the proof of Theorem 1 goes through verbatim (nothing in Lemmas 2–3 or the [7] verification is specific to the modulus 4 beyond χ being real, primitive and nonprincipal), but we have not restated it, so for $q = 3$ — and for every other real primitive character — Theorem 2 is, in this paper, a theorem about the limiting random variable $X_{m,\chi}$ only. (2) Hypotheses. LI is not among the hypotheses of Theorem 2; GRH enters through the reality of the ordinates and the closed forms (8), (10), and $L(\frac{1}{2}, \chi) \neq 0$ (automatic for $q = 3, 4$) through the Hadamard product. (3) Constants. The constants are effective in principle (ρ is a computable supremum, numerically 0.98754; the Riemann–von Mangoldt constants for fixed q are classical) but the large- t bound uses only the zeros in $(M, 2M]$ and is extremely wasteful; no attempt has been made to make $M_0(q)$ small, and at the heights of Table 3 the proved bounds on the tail terms are not yet numerically dominant — the table is a consistency check on the asymptotics, not a certified enclosure. (4) Uniformity. Nothing is claimed uniformly in q ; a uniform version would require the window count and Proposition 3 with explicit q -dependence, which we have not carried out (cf. the q -aspect asymptotics of Fiorilli–Martin [9], whose densities-near- $\frac{1}{2}$ regime is the natural companion statement). (5) The reference densities of Remark 6 remain conditional computations under GRH+LI, as throughout this literature.

5 Methods and verification

Zeros of $L(s, \chi_4)$ and $L(s, \chi_3)$ (511 resp. 537 ordinates to 25 digits) were computed by Hardy Z-scans with derived phases, validated by reality of Z to $\sim 10^{-27}$, Riemann–von Mangoldt counts, and the known first ordinates.

Certified enclosures for the densities

The densities of Table 1 were produced with a tail model: the unseen zeros $\gamma > \Gamma$ ($\Gamma = \gamma_{511} = 639.9157\dots$ for χ_4 , $\gamma_{537} = 699.7747\dots$ for χ_3) were replaced by a Gaussian whose

variance was *estimated* from the zero-counting density. This subsection replaces the estimate by a certified computation and states precisely which error terms are proved and which remain estimated. Throughout, the standing hypotheses are $\text{GRH}(\chi) + \text{LI}(\chi)$ (as for every density in this literature) together with the completeness of the computed ordinate list up to Γ (validated by the Hardy Z -scans and Riemann–von Mangoldt counts described above; the enclosure script repeats a smoothed count check, which matches to 0.6 resp. 1.3 of a zero, within the $S(T)$ fluctuation).

Under LI, $\delta_q(m) = \mathbb{P}(1 + V > 0)$ with $V = \sum_{\gamma > 0} 2a_\gamma(m) \cos \theta_\gamma$, whose characteristic function factors over the computed and unseen ordinates,

$$\Phi(t) = \prod_{\gamma > 0} J_0(2a_\gamma t) = P(t) \Psi(t), \quad P = \prod_{\gamma \leq \Gamma}, \quad \Psi = \prod_{\gamma > \Gamma}.$$

Since $|J_0(x)| \leq \min\{1, \sqrt{2/(\pi x)}\}$ for $x > 0$ (the classical envelope; the extremal value $\sup_{x > 0} \sqrt{x} |J_0(x)| = \sqrt{2/\pi} = 0.79788\dots$ is re-verified numerically in the script, which uses the rounded-up constant 0.8), Φ is integrable, V has a continuous density, and the Gil–Pelaez inversion is an identity:

$$\delta_q(m) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Phi(t) \frac{\sin t}{t} dt. \quad (16)$$

The enclosure controls the three gaps between (16) and what a finite computation can evaluate.

(i) *The tail variance is certified, not estimated.* By Proposition 2,

$$V_T := \sum_{\gamma > \Gamma} 2a_\gamma(m)^2 = 4M \frac{\Lambda'}{\Lambda}(m+1) - \sum_{\gamma \leq \Gamma} 2a_\gamma(m)^2,$$

with no reference to the zero-counting density. The subtracted finite sum inherits the 25-digit accuracy of the ordinates (effect $< 10^{-20}$); the closed form is evaluated by `mpmath` at 40 digits through two independent routes that must agree to 25 digits, anchored at $L(2, \chi_4) = G$ (Catalan), at a directly summed series for $L(2, \chi_3)$, at $(L'/L)(1, \chi_q)$ expressed through generalized Stieltjes constants, and at the closed form for σ_0^2 . This is high-precision floating point, *not* interval arithmetic; we budget a defensive slack of 10^{-15} for it, absorbed into the error radius below. The script aborts if any $V_T \leq 0$; in fact V_T ranges from 0.0030943 ($q = 3$, $m = 0$) to 139.71 ($q = 4$, $m = 100$); see Table 4.

(ii) *The unseen product is a Gaussian with a proved quartic correction.* Since $a_\gamma = 2M/\sqrt{M^2 + \gamma^2}$ is strictly decreasing in γ , every unseen ordinate satisfies $2a_\gamma t \leq 2a_\Gamma t \leq 1$ whenever $t \leq t_{\max} := 1/(2a_\Gamma)$. On $|z| \leq 1$ we use

$$0 \leq -\log J_0(z) - \frac{z^2}{4} \leq c_4 z^4, \quad c_4 = -\log J_0(1) - \frac{1}{4} = 0.0176210647\dots, \quad (17)$$

with the *sharp* constant: from the product $J_0(z) = \prod_n (1 - z^2/j_{0,n}^2)$ over the Bessel zeros, $-\log J_0(z) = \sum_{k \geq 1} (\sigma_k/k) z^{2k}$ with Rayleigh sums $\sigma_k = \sum_n j_{0,n}^{-2k} > 0$, valid for $|z| < j_{0,1} = 2.40\dots$; $\sigma_1 = \frac{1}{4}$ (Rayleigh), and $z^{2k} \leq z^4$ for $k \geq 2$, $|z| \leq 1$, so the middle member of (17) equals $\sum_{k \geq 2} (\sigma_k/k) z^{2k} \in [0, c_4 z^4]$ with $c_4 = \sum_{k \geq 2} \sigma_k/k$, attained at $z = 1$. (The script uses the rounded-up $c'_4 = 0.018$ and re-verifies (17) on a grid.) Summing over the unseen zeros: for $0 \leq t \leq t_{\max}$,

$$\Psi(t) = \exp\left(-\frac{1}{2} V_T t^2 + \varepsilon(t)\right), \quad -c'_4 S_4 t^4 \leq \varepsilon(t) \leq 0, \quad S_4 := \sum_{\gamma > \Gamma} (2a_\gamma)^4 \leq (2a_\Gamma)^2 \cdot 2V_T,$$

the last bound because $(2a_\gamma)^2 \leq (2a_\Gamma)^2$ beyond Γ and $\sum_{\gamma > \Gamma} (2a_\gamma)^2 = 2V_T$. The sign information is exploited: the Gaussian model *overstates* $|\Psi|$, and with $w(t) := c'_4 S_4 t^4$ the factor $e^{\varepsilon(t)} \in [e^{-w}, 1]$ is enclosed by its midpoint,

$$\left| e^{\varepsilon(t)} - \frac{1+e^{-w(t)}}{2} \right| \leq \frac{1-e^{-w(t)}}{2},$$

which halves the tail-model radius relative to the symmetric bound $e^w - 1$.

(iii) *Enclosure.* With $G(t) := e^{-V_T t^2/2}$ and a truncation point $T_1 \leq t_{\max}$,

$$\delta_q(m) = \frac{1}{2} + \frac{1}{\pi} \left(\underbrace{\int_0^{T_1} P G \frac{1+e^{-w}}{2} \frac{\sin t}{t} dt}_{I, \text{ computed}} + R \right), \quad |R| \leq E_1 + E_2,$$

$$E_1 = \int_0^{T_1} |P| G \frac{1-e^{-w}}{2} dt \quad (\text{tail-model radius, from the midpoint enclosure of } e^\varepsilon),$$

$$E_2 = \frac{(0.8)^K}{\sqrt{\prod_{i \leq K} 2a_{\gamma_i}}} \cdot \frac{T_1^{-K/2}}{K/2} \geq \left| \int_{T_1}^\infty \Phi \frac{\sin t}{t} dt \right| \quad (\text{truncation; envelope on the first } K \text{ ordinates,}$$

all other factors bounded by 1, $|\sin t/t| \leq 1/t$). The script chooses $K \in \{24, \dots, 72\}$ and $T_1 \leq t_{\max}$ so that $E_2 \leq \pi \times 10^{-15}$; this holds at every (q, m) computed. What remains *estimated* rather than certified, stated openly: the quadrature error of I and of E_1 (composite Simpson on 2^{17} subintervals in IEEE double precision; a step-doubling Richardson comparison is below 10^{-12} at every (q, m) , and is budgeted as $E_3 = 10^{-12}$; E_1 , itself a bound of size $\lesssim 5 \times 10^{-7}$, is inflated by 50% against its own quadrature error), the double-precision rounding of the 511-fold product ($\lesssim 10^{-13}$ relative), and the 10^{-15} variance slack of (i), together budgeted as $E_4 = 2 \times 10^{-11}$ plus the computed slack sensitivity. The radius of the quoted intervals is $(E_1 + E_2 + E_3 + E_4)/\pi$, dominated everywhere by the proved bound E_1 . These enclosures are rigorous *modulo* exactly the items just listed (double-precision quadrature with Richardson control and the anchored 40-digit evaluation of σ_m^2); no interval arithmetic was used, and we do not claim otherwise.

Corrected values, and what was wrong. The enclosures confirm Table 1 at $(q, m) = (4, 0)$, $(4, 20)$, $(4, 100)$, $(3, 0)$ and move the sixth decimal in the other ten entries by 1–4 units; the corrected six-decimal values are

$$\delta_4 : 0.995928, 0.797630, 0.694521, 0.645851, 0.571870, 0.538455, 0.513778;$$

$$\delta_3 : 0.999063, 0.836880(*), 0.723473, 0.666554, 0.578536, 0.540768, 0.519717(*),$$

(*): two sixth decimals are *not resolved* by the enclosures. At $(3, 1)$ the interval $[0.83688043, 0.83688053]$ straddles the rounding boundary 0.8368805 (the value is 0.836880 or 0.836881; the midpoint 0.83688048 sits within 2.5×10^{-9} of the boundary, so no computation at this zero height can be expected to settle it). At $(3, 60)$ the interval $[0.51971743, 0.51971753]$ straddles 0.5197175 (the value is 0.519717 or 0.519718). All twelve remaining sixth decimals, including $(4, 100)$, are certified by the enclosures. The cause of the drift is quantitatively identified: re-running the certified pipeline with the old smoothed-tail variance in place of V_T reproduces the deviations (e.g. -3.2×10^{-6} of the observed -3.5×10^{-6} at $(3, 1)$, the small remainder being the old code's

q	m	V_T	$\delta_q(m)$ lower	$\delta_q(m)$ upper	width	Tab. 1 – mid
4	0	0.0034836	0.9959278380	0.9959280127	1.7×10^{-7}	+0.07
4	1	0.0313527	0.7976303685	0.7976304296	6.1×10^{-8}	−2.40
4	2	0.0870906	0.6945210810	0.6945211213	4.0×10^{-8}	−1.10
4	3	0.1706968	0.6458510415	0.6458510746	3.3×10^{-8}	−1.06
4	8	1.0067182	0.5718700860	0.5718701185	3.3×10^{-8}	−1.10
4	20	5.8541890	0.5384552785	0.5384553290	5.1×10^{-8}	−0.30
4	100	139.71245	0.5137782134	0.5137783837	1.7×10^{-7}	−0.30
3	0	0.0030943	0.9990630500	0.9990633364	2.9×10^{-7}	−0.19
3	1	0.0278484	0.8368804319	0.8368805308	9.9×10^{-8}	−3.48
3	2	0.0773564	0.7234728791	0.7234729425	6.3×10^{-8}	−2.91
3	3	0.1516181	0.6665542399	0.6665542896	5.0×10^{-8}	−2.26
3	8	0.8942043	0.5785355754	0.5785356134	3.8×10^{-8}	−1.59
3	20	5.2001295	0.5407681867	0.5407682369	5.0×10^{-8}	−1.21
3	60	45.202127	0.5197174290	0.5197175306	1.0×10^{-7}	−1.48

Table 4: Certified enclosures $[\delta_{\text{lo}}, \delta_{\text{hi}}]$ for the logarithmic densities, under GRH + LI and completeness of the ordinate lists to height Γ (511 ordinates for χ_4 , 537 for χ_3). V_T is the certified variance of the unseen-zero tail. The last column is the deviation of the previously printed six-decimal value from the interval midpoint, in units of 10^{-6} ; entries exceeding 1 in absolute value are discrepancies beyond the sixth decimal (ten of the fourteen). All widths are below 3×10^{-7} , against the design target 5×10^{-6} .

coarser quadrature). The classical anchors are unaffected: our intervals contain $0.9959279\dots$ and $0.9990632\dots$ at $m = 0$, consistent with $[1, 9]$ to all six printed decimals.

Correction to Remark 5. That remark asserted that replacing the estimated tail by the exact one moves the densities of Table 1 by less than their quoted precision. The enclosures above show this is false at the sixth decimal for ten of fourteen entries (largest shift 3.5×10^{-6} , at $(q, m) = (3, 1)$); the remark’s claim should be read as “below 5×10^{-6} ”, and Table 4 now supersedes the affected entries.

The two computations, honestly described. The deterministic Gil–Pelaez inversion and the Monte Carlo evaluation of Section 5 share their inputs: the identical 511- resp. 537-ordinate lists and (in their original form) the identical Gaussianized tail. Their mutual agreement (largest deviation 5.5×10^{-4} , i.e. 2.2 binomial standard errors at the actual sample counts of Table 1) therefore validated only the *evaluation* step — quadrature against sampling — and could not have detected an error in the shared tail model. They are independent as implementations, not as models. The enclosures of this subsection close that gap from the model side: the tail variance is now the certified V_T of (i), the non-Gaussianity of the tail is bounded by the proved E_1 , and a Monte Carlo rerun with the certified V_T in the tail agrees with the enclosure midpoints within half a standard error at each tested point (2×10^6 samples at $(q, m) = (4, 0), (4, 1), (3, 1)$).

General real primitive modulus. For use beyond $q \in \{3, 4\}$ we record the general form of Theorem 1 and of the variance identity. Let χ be a real primitive character mod q with

$\chi(-1) = (-1)^a$, $a \in \{0, 1\}$, and assume $\text{GRH}(\chi)$, $\text{LI}(\chi)$, and the no-central-zero hypothesis

$$\text{NCZ}(\chi) : \quad L(\tfrac{1}{2}, \chi) \neq 0.$$

Define $R_m(x) = -\sum_{p \leq x} \chi(p) p^m \log p$ (nonresidues minus residues), $E_m = (2m+1)x^{-(m+1/2)} R_m$, and $a_\gamma(m) = 2M/\sqrt{M^2 + \gamma^2}$ over the positive ordinates of $L(s, \chi)$. Then parts (i)–(iii) of Theorem 1 hold verbatim with $C(\chi) = \sum_{\gamma_\chi > 0} 2\gamma_\chi^{-2}$, and the variance identity becomes, with $\xi(s, \chi) = (q/\pi)^{(s+a)/2} \Gamma(\frac{s+a}{2}) L(s, \chi)$,

$$\sigma_m^2(\chi) = 4M \frac{\xi'}{\xi}(m+1, \chi), \quad C(\chi) = (\log \xi)''(\tfrac{1}{2}, \chi).$$

What changes and what does not. *Unchanged*: the prime-square bias term — the $k = 2$ prime powers contribute $\sum_{p \leq \sqrt{x}, p \nmid q} p^{2m} \log p = x^{m+1/2}/(2m+1) + o(x^{m+1/2})$ because $\chi(p)^2 = 1$ for every $p \nmid q$ and the finitely many $p \mid q$ contribute $O_m(x^m \log x)$, so the normalized mean is 1 for every real χ ; the Chernoff/Chebyshev bounds of (iii), which use only the arcsine moment generating function. *Changed*: the gamma factor and conductor in ξ (parity a enters the variance asymptotic only through the constant term $2a - 1$ of Proposition 3); all implied constants of the explicit-formula lemmas, which now depend on q , and the first-ordinate constants ($|\rho + m| \geq \gamma_1(\chi)$ in Lemma 2; numerically $\gamma_1(\chi_4) = 6.0209\dots$, $\gamma_1(\chi_3) = 8.0397\dots$, and $\gamma_1(\chi) > 0$ is exactly NCZ under GRH); for $a = 0$ the pole of $1/s$ in the Perron contour meets the trivial zero of $L(s, \chi)$ at $s = 0$, and the classical formula carries an extra $-\log x + b(\chi)$ term [17, Ch. 19], absorbed by the $O(\log x)$ floor of (2); and the functional equation input, $\xi(s, \chi) = \xi(1-s, \chi)$, i.e. root number $+1$, which holds for every real primitive χ by the evaluation of the sign of the quadratic Gauss sum ($\tau(\chi) = \sqrt{q}$ for $a = 0$, $i\sqrt{q}$ for $a = 1$; [17, Ch. 2] for prime modulus, the general case reducing to it via the factorization of Gauss sums over coprime moduli — for the moduli 3, 4 used here the two-term sums are evaluated directly in the verification script). On the no-central-zero hypothesis itself: for χ_4 it is the alternating series $L(\frac{1}{2}, \chi_4) = \sum_k (-1)^k (2k+1)^{-1/2} > 0$; for χ_3 it is *not* an alternating-series statement (the zero terms $\chi_3(3k) = 0$ interleave), but it is still unconditional: the Dirichlet series $\sum \chi_3(n) n^{-1/2}$ converges by Dirichlet's test (bounded character sums) and equals $L(\frac{1}{2}, \chi_3)$ by analytic continuation, and regrouping the convergent series in blocks of three gives

$$L(\tfrac{1}{2}, \chi_3) = \sum_{k \geq 0} \left[\frac{1}{\sqrt{3k+1}} - \frac{1}{\sqrt{3k+2}} \right] \geq 1 - \frac{1}{\sqrt{2}} > 0$$

(every block positive; numerically $0.4808675\dots$). For general q we do not prove NCZ: under GRH one has $L(\frac{1}{2}, \chi) \geq 0$ (a negative central value would force a real zero in $(\frac{1}{2}, 1)$), so NCZ is then exactly central nonvanishing, which is Chowla's conjecture for quadratic characters — known in every numerically checked case but open in general; it must be assumed, or verified computationally for the modulus at hand, as part of the hypotheses. Nothing in this paper uses NCZ beyond $q \in \{3, 4\}$, where it is proved above. For the record, the certified constants at the two moduli used here are

$$\begin{aligned} C(\chi_4) &= 0.15602964988964488\dots, & C(\chi_3) &= 0.11340215685275857\dots, \\ \sigma_0^2(\chi_4) &= 0.15556797992358593\dots, & \sigma_0^2(\chi_3) &= 0.11322996985747234\dots \end{aligned}$$

Exact races: segmented sieves to 10^{10} with exact signed 128-bit (integer weights) or compensated long-double (fractional weights) accumulation, cross-checked against independent implementations at overlapping ranges; the $m = 0$ anchors (first flip 26,861; no mod-3 flip below 10^{10}) reproduce the classical values. The finite- x logarithmic measure of a set S reported in the introduction is $(\log(X/3))^{-1} \int_3^X \mathbf{1}_S(t) dt/t$ for modulus 4 (the race is tracked from the first odd prime) and $(\log(X/2))^{-1} \int_2^X \mathbf{1}_S(t) dt/t$ for modulus 3 (the race starts at $p = 2$); intervals on which $D_m = 0$ are assigned to the previous leader. The convention is not cosmetic: at $X = 10^{10}$ the choice of lower endpoint alone shifts the normalization by $\log 3 / \log X \approx 4.8\%$. The density kernel is derived for the log-weighted (θ -form) race; partial summation gives every term of the unlogged race the same $1/\log x$ factor, which cancels under the bias normalization, so the limiting distribution and density coincide — the same passage Rubinstein–Sarnak make at $m = 0$. The explicit-formula kernel derives from a uniform smoothed-Perron framework developed in a companion manuscript (analytic computation of $S_m(x)$ from zeros, exact to 10^9 at $m = 1$), available with all code and data at <https://github.com/raghavs1729/weighted-prime-races> (archived at doi:10.5281/zenodo.21947102). Every headline number in this note was re-computed by a second implementation. For the sieve computations the two implementations are independent throughout; for the densities they shared the ordinate lists and, prior to the enclosures above, the Gaussian tail model, so their agreement tested evaluation error only — the tail model itself is certified by Table 4.

Assumptions and limitations. Densities are conditional on GRH + LI, as throughout this literature. The existence of the limiting distribution, the continuity of $m \mapsto \delta(m)$, and the strict bounds $\frac{1}{2} < \delta(m) < 1$ are proved for all $m \in (-\frac{1}{2}, \infty)$ in Section 2 (the $m > 0$ part is the routine direction of the extension: the only changes are the $(1 + x^m) \log x$ error floor in the explicit formula and the constant $2 \max K + 1$ in the continuity argument). The dissolution law, heuristic in earlier drafts, is proved with an error term in Theorem 2 (for the density of the limiting variable; the transfer to the race itself uses LI and is stated for $q = 4$). The finite- x log-measure for $(q, m) = (3, 2)$ shows the weakest agreement in the study (0.023 at 10^{10}), plausibly a slow almost-periodic phase; we flag it rather than explain it.

Outlook: growing weights and short intervals. Everything in this note keeps m fixed as $x \rightarrow \infty$. In the joint limit $m = m(x) \rightarrow \infty$ the weight localizes the race: writing $p = x(1 - u)$, the weight $p^m = x^m e^{m \log(1-u)}$ is within a bounded factor of its maximum only for $u \ll 1/m$, so the fluctuating part of $\sum_{p \leq x} \chi(p) p^m \log p$ is dominated by the primes in the short window $x - p \ll x/m$. A growing-weight race is in this sense a smoothed race between primes in a short interval at the edge of the range, and the weight aspect should connect to the extensively studied statistics of primes in short intervals — in particular to the Gaussian limit laws, under RH and a quantitative form of LI, of Leung [22] for primes in multiple short intervals, which include short-interval race applications. We state this only as a direction: we prove nothing in any joint regime $m = m(x) \rightarrow \infty$, and even the existence of a limiting distribution there is open.

References

- [1] M. Rubinstein, P. Sarnak, *Chebyshev’s bias*, Experiment. Math. **3** (1994), 173–197.
- [2] M. Aoki, S. Koyama, *Chebyshev’s bias against splitting and principal primes in global fields*, J. Number Theory **245** (2023), 233–262. arXiv:2203.12266.

- [3] A. Sheth, *Euler products at the centre and applications to Chebyshev's bias*, Math. Proc. Cambridge Philos. Soc. **179** (2025), 331–349. arXiv:2405.01512.
- [4] K. Shimada, S. Koyama, *Weighted prime number theorem on arithmetic progressions with refinements*, Mathematics **13** (2025), no. 16, 2564.
- [5] P. Humphries, *The distribution of weighted sums of the Liouville function and Pólya's conjecture*, J. Number Theory **133** (2013), 545–582. arXiv:1108.1524.
- [6] A. Hamieh, H. Kadiri, G. Martin, N. Ng, *Comparative prime number theory problem list*, arXiv:2407.03530 (2024), Problem 12.
- [7] A. Akbary, N. Ng, M. Shahabi, *Limiting distributions of the classical error terms of prime number theory*, Q. J. Math. **65** (2014), 743–780.
- [8] L. Devin, *Chebyshev's bias for analytic L -functions*, Math. Proc. Cambridge Philos. Soc. **169** (2020), 103–140. arXiv:1706.06394.
- [9] D. Fiorilli, G. Martin, *Inequities in the Shanks–Rényi prime number race: an asymptotic formula for the densities*, J. reine angew. Math. **676** (2013), 121–212.
- [10] X. Meng, *Chebyshev's bias for products of k primes*, Algebra Number Theory **12** (2018), 305–341.
- [11] A. Languasco, A. Zaccagnini, *Computing the Mertens and Meissel–Mertens constants for sums over arithmetic progressions*, Experiment. Math. **19** (2010), 279–284.
- [12] D. Fiorilli, *Highly biased prime number races*, Algebra Number Theory **8** (2014), 1733–1767.
- [13] Y. Lamzouri, *A bias in Mertens' product formula*, Int. J. Number Theory **12** (2016), 97–109. arXiv:1410.3777.
- [14] M. Hayani, *Chebyshev's bias without linear independence*, arXiv:2512.23302 (2025).
- [15] D. J. Platt, T. S. Trudgian, *On the first sign change of $\theta(x) - x$* , Math. Comp. **85** (2016), 1539–1547. arXiv:1407.1914.
- [16] M. Aymone, *A note on prime number races and zero free regions for L functions*, Int. J. Number Theory **18** (2022). arXiv:2001.00764.
- [17] H. Davenport, *Multiplicative Number Theory*, 3rd ed., revised by H. L. Montgomery, Graduate Texts in Mathematics **74**, Springer, New York, 2000.
- [18] H. L. Montgomery, *The zeta function and prime numbers*, Proceedings of the Queen's Number Theory Conference (Kingston, Ont., 1979), Queen's Papers in Pure and Appl. Math. **54** (1980), 1–31.
- [19] K. Yamamuro, *A weighted prime number race through some perspective of probability theory*, Ann. Univ. Sci. Budapest. Sect. Comput. **58** (2025), 299–313.
- [20] M. Hayani, *On the role of higher roots in prime ideal races*, arXiv:2607.23150 (2026).

- [21] A. Bailleul, M. Hayani, T. Untrau, *A Wasserstein metric approach to generalized Skewes' numbers. I. Prime number races*, arXiv:2603.20093 (2026).
- [22] S.-K. Leung, *Joint distribution of primes in multiple short intervals*, arXiv:2401.04000 (2024).
- [23] G. Martin, N. Ng, *Inclusive prime number races*, Trans. Amer. Math. Soc. **373** (2020), no. 5, 3561–3607. arXiv:1710.00088.
- [24] Z. W. Birnbaum, *On random variables with comparable peakedness*, Ann. Math. Statist. **19** (1948), 76–81.
- [25] F. Proschan, *Peakedness of distributions of convex combinations*, Ann. Math. Statist. **36** (1965), 1703–1706.
- [26] W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. II, 2nd ed., Wiley, New York, 1971.